

M13 Batch 6 — Full Set

M13.8 Instruments (ATA 31) · PDF pages 352–491

General properties, pressure measuring devices, pitot-static systems and their faults, altimeters, airspeed indicators, machmeters and VSIs.

416 questions · Level 3: 122 · Level 2: 213 · Level 1: 81

● marks a question built on a fact you highlighted in the manual (370 of 416). Topics run in manual page order; inside each topic the questions are grouped by level, and the answers with explanations follow at the end of that topic.

No calculations, no unit conversions, no worked examples, nothing type-specific.

Classification

PDF pages 352–352 · 13 questions

LEVEL 1 — RECALL

1. To which two subgroups can every instrument of the classical cockpit be assigned ●
 - a. Airframe monitoring and aircraft monitoring
 - b. Flight monitoring and aircraft monitoring
 - c. Flight monitoring and engine monitoring
2. To which group does the artificial horizon indicator ADI belong ●
 - a. Aircraft monitoring
 - b. Flight monitoring
 - c. Airframe monitoring

LEVEL 2 — UNDERSTAND AND EXPLAIN

3. Which group of instruments does the standby set of a modern cockpit duplicate ●
 - a. engine monitoring
 - b. airframe monitoring
 - c. flight monitoring
4. An exhaust gas temperature indicator is fitted in a classical cockpit. Which group and subdivision does it belong to? ●
 - a. Aircraft monitoring, as an engine monitoring device
 - b. Aircraft monitoring, as an airframe monitoring device
 - c. Flight monitoring, as a pressure monitoring device
5. Into which two kinds of devices is aircraft monitoring divided ●
 - a. Airframe monitoring devices and standby devices
 - b. Flight monitoring devices and engine monitoring devices
 - c. Airframe monitoring devices and engine monitoring devices
6. Why do the aircraft monitoring systems of the modern cockpit carry different names such as EICAS or ECAM ●
 - a. Because the name depends on the display size
 - b. Because the name depends on the flight phase
 - c. Because the name depends on the aircraft type

7. Which task do EICAS and ECAM perform in the modern cockpit ●
 - a. They perform flight monitoring
 - b. They perform aircraft monitoring
 - c. They perform maintenance data recording
8. Why is the CMS counted as part of aircraft monitoring ●
 - a. Because it delivers information to the standby instruments
 - b. Because it delivers information to the maintenance personnel
 - c. Because it delivers information to the flight crew
9. The electronic displays of a glass cockpit are lost in flight. Which speed and attitude information remains available from the standby set? ●
 - a. Airspeed from the standby airspeed indicator and heading from the standby horizon
 - b. Mach number from the standby machmeter and attitude from the standby horizon
 - c. Airspeed from the standby airspeed indicator and attitude from the standby horizon
10. A fuel pressure indication changes during flight. Which value is the instrument responding to?
 - a. Pressure of fluids
 - b. Earth's magnetism
 - c. Pressure of air

LEVEL 3 — APPLY AND REASON

11. An aircraft presents its altimeter, airspeed indicator and artificial horizon indicator on electronic displays. To which system of the modern cockpit are these instruments assigned? ●
 - a. CMS (Central Maintenance System)
 - b. EICAS (Engine Indicating and Crew Alerting System)
 - c. EFIS (Electronic Flight Instrument System)
12. The standby devices of a modern cockpit cover all but one of the flight parameters shown on the PFD. Which parameter is left without a standby device ●
 - a. altitude
 - b. vertical speed
 - c. airspeed
13. An engineer needs aircraft system information for troubleshooting rather than an indication meant for the crew. Which aircraft monitoring system delivers it? ●
 - a. The EFIS (Electronic Flight Instrument System)
 - b. The ECAM (Electronic Centralized Aircraft Monitoring)
 - c. The CMS (Central Maintenance System)

Answers — Classification

1. **b** The text splits classical cockpit instruments into flight monitoring and aircraft monitoring. (L1, PDF p. 352)
2. **b** The ADI is listed among the flight monitoring instruments. (L1, PDF p. 352)
3. **c** Altimeter, airspeed indicator, horizon and compass are all listed as flight monitoring instruments. (L2, PDF p. 352)
4. **a** Exhaust gas temperature indicators are named as engine monitoring devices under aircraft monitoring. (L2, PDF p. 352)
5. **c** Aircraft monitoring covers airframe monitoring devices and engine monitoring devices. (L2, PDF p. 352)
6. **c** The text states that aircraft monitoring instruments have different names depending on aircraft type. (L2, PDF p. 352)
7. **b** EICAS and ECAM are the names given to the aircraft monitoring instruments. (L2, PDF p. 352)
8. **b** The CMS is used for aircraft monitoring because it delivers information to maintenance personnel. (L2, PDF p. 352)

- 9. c The standby set contains an airspeed indicator and a horizon, but no machmeter. (L2, PDF p. 352)
- 10. a Fluid pressure is listed separately from air pressure as a value that operates instruments. (L2, PDF p. 352)
- 11. c All three named instruments are listed as flight monitoring instruments, and flight monitoring instruments go to the EFIS. (L3, PDF p. 352)
- 12. b The standby set holds altimeter, airspeed indicator, horizon and compass, so of the PFD parameters only vertical speed has no standby device. (L3, PDF p. 352)
- 13. c The CMS is the aircraft monitoring system that delivers information to the maintenance personnel. (L3, PDF p. 352)

Cockpit displays and the Basic T

PDF pages 366–366 · 6 questions

LEVEL 1 — RECALL

14. A panel layout is described as Basic T. Which four instruments form that T? ●
- Airspeed indicator, attitude indicator, altimeter and compass
 - Attitude indicator, altimeter, compass and turn coordinator
 - Airspeed indicator, attitude indicator, altimeter and vertical speed indicator
15. Where is the attitude indicator located in the Basic T configuration ●
- In the middle at the top
 - In the middle at the bottom
 - On the left at the top

LEVEL 2 — UNDERSTAND AND EXPLAIN

16. Of the two combined indicators, which one adds the flight director indication and which one adds radio navigation indications ●
- the ADI and the magnetic standby compass respectively
 - the HSI and the ADI respectively
 - the ADI and the HSI respectively
17. A pilot reads his magnetic heading and a VOR course deviation from one round instrument. Which indicator is he using? ●
- The ADI
 - The variometer
 - The HSI
18. A glass cockpit aircraft has no separate round flight instruments. Which statement describes the arrangement of the flight parameters? ●
- They are laid out on the PFD and correspond to the Basic T configuration
 - They are laid out on the PFD in a configuration that replaces the Basic T
 - They are laid out on the navigation display in Basic T configuration

LEVEL 3 — APPLY AND REASON

19. An instrument is to be fitted in the middle at the top of a Basic T panel and must also display the indication from the flight director system. Which indicator is required? ●
- The horizontal situation indicator HSI
 - The attitude director indicator ADI
 - The turn coordination indicator

Answers — Cockpit displays and the Basic T

14. a The four instruments arranged in the T shape are the airspeed indicator, attitude indicator, altimeter and compass; turn coordination and vertical speed sit below them. (L1, PDF p. 366)
15. a The Basic T places the attitude indicator centrally at the top of the group. (L1, PDF p. 366)
16. c The ADI adds the flight director indication to the artificial horizon; the HSI adds radio navigation indications to the compass. (L2, PDF p. 366)
17. c The HSI is the instrument that combines compass and radio navigation information. (L2, PDF p. 366)
18. a The PFD keeps the Basic T arrangement rather than abandoning it. (L2, PDF p. 366)

- 19. b** The middle top position of the Basic T carries the attitude indicator, and the ADI is the attitude indicator that also shows the flight director indication. (L3, PDF p. 366)

The atmosphere and ISA

PDF pages 370–374 • 17 questions

LEVEL 1 — RECALL

20. What is the proportion of nitrogen in dry, clean air directly above the surface of the earth ●
- 78%
 - 1%
 - 21%
21. What is the proportion of oxygen in dry, clean air directly above the surface of the earth ●
- 78%
 - 21%
 - 11%
22. An air data system is being described to a student. Which two measurements does flight monitoring require? ●
- Static pressure and dynamic pressure
 - Static pressure and air density
 - Dynamic pressure and air temperature
23. What is the other name for dynamic pressure ●
- Ram pressure
 - Atmospheric pressure
 - Barometric pressure

LEVEL 2 — UNDERSTAND AND EXPLAIN

24. How does the temperature behave with increasing height in the troposphere ●
- It decreases by 6.5 °C every 100 metres
 - It decreases by 0.65 °C every 100 metres
 - It increases by 0.65 °C every 100 metres
25. Which air data measurement is also called atmospheric pressure and which one is also called ram pressure ●
- the dynamic pressure and the static pressure respectively
 - the static pressure and the total pressure respectively
 - the static pressure and the dynamic pressure respectively
26. Which two indications become possible once temperature is applied as a corrective value ●
- true airspeed and static air temperature
 - barometric altitude and true airspeed
 - airspeed and vertical speed
27. How do static pressure, temperature and density behave as altitude increases ●
- All three become higher
 - Only temperature becomes lower
 - All three become lower

28. True airspeed and static air temperature are to be indicated as well. Which value must be applied as a corrective? ●
- Temperature
 - Density
 - Humidity
29. A pneumatic instrument is calibrated in the workshop. Which values form the basis of that calibration?
- The standard atmosphere values at mean sea level
 - The actual airfield values at the time of installation
 - The tropopause values of the standard atmosphere
30. Why has the international standard atmosphere been established
- To predict the weather conditions along a planned route
 - To compare flight performance under greatly differing conditions
 - To set the pressure scale of the altimeter before every flight

LEVEL 3 — APPLY AND REASON

31. A crew flies at a level that lies inside the troposphere over the equator but above it over the poles. Which printed thicknesses account for this? ●
- About 16.8 km at the equator and about 8.5 km at the poles
 - About 8.5 km at the equator and about 16.8 km at the poles
 - About 16.8 km at the equator and about 11 km at the poles
32. An aircraft climbs and its static pressure falls. Which indication does a mechanical air data instrument derive from that falling pressure? ●
- the barometric altitude
 - the static air temperature
 - the indicated airspeed
33. An air data system has static and dynamic pressure available but no temperature input. Which indication is it unable to produce? ●
- the vertical speed
 - the barometric altitude
 - the true airspeed
34. An airspeed indicator with bellows works on the difference between total and static pressure. Of the two air data measurements required for flight monitoring, which one does it use for this? ●
- the barometric altitude
 - the static pressure
 - the dynamic pressure
35. In cruise above the transition altitude all pilots set 1013.2 hPa on the altimeter. Which standard atmosphere value is this setting taken from? ●
- The pressure at the tropopause
 - The pressure at mean sea level
 - The pressure at the transition level
36. How does the air temperature of the standard atmosphere behave with increasing height directly above the tropopause
- It begins to rise with height
 - It continues to fall at the same rate
 - It stops falling and stays constant

Answers — The atmosphere and ISA

- 20. a Nitrogen makes up 78% of dry clean air at the surface. (L1, PDF p. 370)
- 21. b Oxygen makes up 21% of dry clean air at the surface. (L1, PDF p. 370)
- 22. a Flight monitoring is built on the two pressures, static and dynamic. (L1, PDF p. 372)
- 23. a Dynamic pressure is the ram pressure produced by the aircraft's motion. (L1, PDF p. 372)
- 24. b The troposphere lapse rate is 0.65 °C per 100 m of climb. (L2, PDF p. 370)
- 25. c The manual gives atmospheric pressure as the other name for static pressure and ram pressure as the other name for dynamic pressure. (L2, PDF p. 372)
- 26. a Temperature as a corrective value makes true airspeed and static air temperature available. (L2, PDF p. 372)
- 27. c All three listed values fall with increasing altitude. (L2, PDF p. 372)
- 28. a Temperature is the corrective value that makes TAS and SAT available. (L2, PDF p. 372)
- 29. a Calibration is referred to the ISA mean sea level values. (L2, PDF p. 372)
- 30. b The ISA exists so performance figures can be compared on a common basis. (L2, PDF p. 374)
- 31. a The troposphere reaches much higher at the equator than at the poles, so the same level can be inside it in one place and above it in the other. (L3, PDF p. 370)
- 32. a Static pressure falls as altitude increases, and it is the static pressure that a mechanical instrument uses for barometric altitude. (L3, PDF p. 372)
- 33. c Barometric altitude, vertical speed and airspeed come from the two pressures alone, while true airspeed needs temperature as a corrective value. (L3, PDF p. 372)
- 34. c Dynamic pressure is the difference between total and static pressure, so the bellows instrument is working on the dynamic pressure of the two measurements. (L3, PDF p. 372)
- 35. b The standard atmosphere fixes the sea level pressure at 1013,25 hPa, which is the pressure the crew sets for flight levels. (L3, PDF p. 374)
- 36. c Identical temperatures at both levels show the lapse rate has stopped at the tropopause. (L3, PDF p. 374)

Altitude, altimeter settings and RVSM

PDF pages 376–384 • 44 questions

LEVEL 1 — RECALL

37. What does HUD stand for ●
 - a. Head Up Display
 - b. Head Unit Display
 - c. Heading Up Display
38. Where is the reference level set on the altimeter ●
 - a. On the vertical speed scale
 - b. On the altitude drum
 - c. On the pressure scale
39. What is elevation ●
 - a. The altitude of the aircraft above mean sea level
 - b. The altitude of an object on the ground above mean sea level
 - c. The altitude of the aircraft above the ground
40. An operator prepares its fleet for RVSM operations. What does this abbreviation mean? ●
 - a. The doubling of vertical safety distances at higher flight altitudes
 - b. The halving of lateral safety distances at higher flight altitudes
 - c. The halving of vertical safety distances at higher flight altitudes
41. What is RVSM airspace ●
 - a. Airspace above FL 290 with 2000 ft vertical separation
 - b. Airspace above FL 290 with 1000 ft vertical separation
 - c. Airspace above FL 285 with 500 ft vertical separation
42. Who must receive additional mandatory training for the introduction of RVSM ●
 - a. Maintenance personnel and ground handlers
 - b. Pilots and maintenance personnel
 - c. Pilots and air traffic controllers

LEVEL 2 — UNDERSTAND AND EXPLAIN

43. Where can the conversion from pressure to indication take place ●
 - a. In the pitot probe itself or separately in the instrument
 - b. In the instrument itself or separately in air data calculators
 - c. In the instrument itself or separately in the transponder
44. Two aircraft show the same air data but the instruments work differently inside. On what does the way they function depend? ●
 - a. On the manufacturer and the reference pressure selected
 - b. On the manufacturer and the state of technological development
 - c. On the operator and the altitude monitoring programme
45. In a modern indicating system, how is all the air data information presented ●
 - a. on one screen called the HUD
 - b. on separate electrical instruments
 - c. on one screen called the PFD

46. How does the HUD present the PFD data ●
- a. So that the pilot can see it only when the autopilot is engaged
 - b. So that the pilot can see it when he looks down at the centre pedestal
 - c. So that the pilot can see it when he looks through the windshield
47. Which of the two transition terms applies prior to landing ●
- a. the transition altitude
 - b. the flight level
 - c. the transition level
48. A barometric altimeter is set to QFE while the aircraft stands on the airfield. What does it indicate? ●
- a. Zero altitude
 - b. Airfield elevation
 - c. Current flight level
49. What does a barometric altimeter set to QFE indicate in flight ●
- a. The altitude above mean sea level, as long as the air pressure does not change
 - b. The flight level, as long as the air pressure does not change
 - c. The height above the airfield, as long as the air pressure does not change
50. An aircraft stands on an airfield with QNH set on its altimeter. What does the altimeter show? ●
- a. Airfield elevation
 - b. Transition level
 - c. Zero altitude
51. A crew selects QNH on both altimeters. In which phases of flight is this setting used? ●
- a. During landing and take-off, above transition altitude and above transition level
 - b. During cruise, above transition altitude and above transition level
 - c. During landing and take-off, up to transition altitude and below transition level
52. Above which altitude do all pilots set the standard pressure in cruise flight ●
- a. 5000 ft above mean sea level
 - b. 10000 ft above mean sea level
 - c. 2000 ft above mean sea level
53. All aircraft in cruise use the same standard pressure setting. Which purpose does this serve? ●
- a. It ensures correct indication of the airfield elevation
 - b. It ensures safe vertical separation
 - c. It ensures a correct height above the airfield
54. What is the transition altitude ●
- a. Prior to landing, the height at which the altimeter is set from QNH to STD
 - b. After take-off, the height at which the altimeter is set from QNH to STD
 - c. After take-off, the height at which the altimeter is set from STD to QNH
55. What is the transition level ●
- a. Prior to landing, the last usable flight level at which the altimeter is set from STD to QNH
 - b. After take-off, the last usable flight level at which the altimeter is set from STD to QNH
 - c. Prior to landing, the last usable flight level at which the altimeter is set from QNH to STD

56. What is the purpose of reducing the vertical separation minimum ●
- a. To reduce the workload of the air data computers
 - b. To accommodate more aircraft in the same airspace
 - c. To allow aircraft to climb above the tropopause
57. Why was the separation above FL 285 formerly increased
- a. Lack of radar coverage and reduced transponder range at high altitude
 - b. Lack of measurement precision and reduced altimeter sensitivity at high altitude
 - c. Lack of autopilot authority and reduced control response at high altitude
58. Besides the pitot-static sensors, the air data computers, the altimeters and the autopilot in altitude hold mode, which further item is listed as RVSM-sensitive? ●
- a. The transponder antenna and its surroundings
 - b. The radio altimeter antennas and their surroundings
 - c. The structure and surroundings of the pitot-static sensors
59. What is the function of the warning device required for flight in RVSM airspace ●
- a. It indicates when the aircraft enters RVSM airspace
 - b. It indicates when the altimeter setting differs from 1013.2 hPa
 - c. It indicates when the aircraft leaves the predetermined height
60. Which autopilot capability is required for flight in RVSM airspace ●
- a. It must be able to maintain the heading
 - b. It must be able to maintain the airspeed
 - c. It must be able to maintain the altitude
61. Which transponder requirement applies to an aircraft flying in RVSM airspace? ●
- a. It must transmit heading and be connected to the altitude measurement unit
 - b. It must transmit altitude and be connected to the altitude measurement unit
 - c. It must transmit altitude and be connected to the radio altimeter
62. Which operational advantage is gained by an RVSM approved aircraft ●
- a. It can use the increased number of flight levels to fly above the tropopause
 - b. It can use the increased number of flight levels to save time and fuel
 - c. It can use a reduced number of flight levels to save maintenance costs
63. A workshop is being prepared for the maintenance of RVSM approved aircraft. Which of the mandatory training subjects concerns the accuracy of the workshop's own equipment? ●
- a. The configuration of air data systems
 - b. Improved precision of test equipment
 - c. Improved altimeter precision
64. A fleet of six aircraft is to be cleared for RVSM. Who must apply for the Airworthiness Approval and for what? ●
- a. The aircraft operator, for each individual aircraft
 - b. The aircraft manufacturer, for each individual aircraft
 - c. The aircraft operator, for the whole fleet at once
65. What does the requirement for RVSM Airworthiness Approval mean for each individual aircraft
- a. Modification according to required service bulletins and a full static system overhaul
 - b. Replacement of the air data computers and the altitude monitoring programme
 - c. Modification according to required service bulletins and the altitude monitoring programme

66. Which detail of the air data computer must the mandatory RVSM training cover ●
- a. Its cooling airflow
 - b. Its part numbers
 - c. Its software loading procedure

LEVEL 3 — APPLY AND REASON

67. An aircraft has electrical air data instruments fed from separate air data calculators. In this fit, where does the conversion from pressure by mechanical means take place ●
- a. in the pitot and static probes
 - b. separately from the instrument
 - c. in the instrument itself
68. One screen in the cockpit carries all the air data information and also every Basic T parameter. Which screen is this ●
- a. the PFD (Primary Flight Display)
 - b. the HUD (Head Up Display)
 - c. the standby horizon
69. Two altimeters on the same parked aircraft are read at the airfield, one showing zero and the other showing the airfield elevation. Which settings are selected? ●
- a. QFE on the first and QNE on the second
 - b. QFE on the first and QNH on the second
 - c. QNH on the first and QFE on the second
70. An aircraft climbs through the transition altitude after take-off. Which change of altimeter setting is required? ●
- a. From QNH to STD (QNE)
 - b. From QFE to QNH
 - c. From STD (QNE) to QNH
71. An aircraft descends through the transition level on the approach. Which change of altimeter setting is required? ●
- a. From QNH to QFF
 - b. From STD (QNE) to QNH
 - c. From QNH to STD (QNE)
72. The low range radio altimeter is unserviceable and the crew needs the height above the airfield. Which setting gives it on the barometric altimeter, and under which condition? ●
- a. QNH, in as far as there is no change in air pressure
 - b. QFE, in as far as there is no change in air pressure
 - c. QFE, in as far as the aircraft stays below the transition altitude
73. An aircraft in cruise above the transition altitude keeps QNH set instead of the standard pressure. What is no longer ensured ●
- a. the indication of the airfield elevation
 - b. the measurement of the height above the ground
 - c. the safe vertical separation from other aircraft

74. An aircraft cruises at 4000 ft above mean sea level in an area with a transition altitude of 5000 ft. Which altimeter setting must be used? ●
- QNE
 - QNH
 - QFF
75. RVSM airspace lies above FL 290 and is therefore flown on flight levels. What must the altimeter be set to there ●
- the standard setting of 1013.2 hPa
 - the QFE of the departure airfield
 - the QNH of the destination airfield
76. An aircraft is cleared to a level above FL 290. Which vertical separation applies there and what is the airspace called? ●
- 1000 ft separation, known as MNPS airspace
 - 1000 ft separation, known as RVSM airspace
 - 2000 ft separation, known as RVSM airspace
77. An aircraft has one altimeter, an autopilot that can hold the altitude, a height deviation warning and a transponder with altitude transmission. Which RVSM equipment requirement does it fail to meet ●
- two independent altimeters
 - an autopilot for altitude hold
 - a transponder with altitude transmission
78. An autopilot can no longer hold the selected altitude. Which requirement for flight in RVSM airspace is no longer met? ●
- A device for indicating when leaving the predetermined height
 - A transponder with altitude transmission
 - An autopilot that can maintain the altitude
79. A dent is found in the skin close to a static port of an RVSM approved aircraft. Which consequence must be expected?
- A considerable influence on the altitude indication
 - A considerable influence on the mach number only
 - A considerable influence on the radio altimeter indication
80. An altimeter fails in cruise and the aircraft loses its RVSM capability. Which action is required? ●
- A change of the altimeter setting to QNH
 - A descent to the transition level
 - A quick exit of RVSM airspace

Answers — Altitude, altimeter settings and RVSM

37. a HUD is short for Head Up Display. (L1, PDF p. 378)
38. c The reference is entered on the altimeter's pressure scale, and the indication follows from it. (L1, PDF p. 380)
39. b Elevation refers to something on the ground measured from mean sea level. (L1, PDF p. 380)
40. c Reduced vertical separation minimum halves the vertical spacing used up high. (L1, PDF p. 382)
41. b RVSM airspace starts above FL 290 and uses 1000 ft spacing. (L1, PDF p. 382)
42. b Training covers both flight crew and maintenance staff. (L1, PDF p. 384)
43. b Either the instrument does the whole job or a separate air data calculator does the conversion. (L2, PDF p. 376)
44. b The manual attributes the differences to manufacturer and technology level. (L2, PDF p. 376)

45. c Modern indicating systems gather all the air data information onto the PFD. (L2, PDF p. 378)
46. c The HUD projects the PFD information into the pilot's forward view. (L2, PDF p. 378)
47. c The transition altitude belongs to the climb after take-off and the transition level to the approach prior to landing. (L2, PDF p. 380)
48. a With QFE set the reading starts from the airfield, so it is zero on the ground. (L2, PDF p. 380)
49. c QFE gives height above the airfield, valid while the pressure stays as set. (L2, PDF p. 380)
50. a QNH refers the altimeter to mean sea level, so on the ground it reads the airfield elevation. (L2, PDF p. 380)
51. c QNH belongs to the low level phases, below the transition altitude or level. (L2, PDF p. 380)
52. a The transition altitude quoted here is 5000 ft MSL. (L2, PDF p. 380)
53. b A common reference means all altimeters err alike, which keeps the vertical spacing valid. (L2, PDF p. 380)
54. b The transition altitude is the climb-out point for changing to the standard setting. (L2, PDF p. 380)
55. a The transition level is the descent point for changing back to QNH. (L2, PDF p. 380)
56. b More usable levels mean more aircraft fit into the same volume of airspace. (L2, PDF p. 382)
57. b Altimeters become less precise with height, so a bigger buffer was needed. (L2, PDF p. 382)
58. c The list of RVSM-sensitive components closes with the structure and surroundings of the pitot-static sensors. (L2, PDF p. 382)
59. c The device warns the crew that the selected altitude is being left. (L2, PDF p. 382)
60. c RVSM depends on the autopilot holding the selected altitude accurately. (L2, PDF p. 382)
61. b The transponder must report altitude taken from the altitude measurement unit. (L2, PDF p. 382)
62. b More available levels give better routings, saving time and fuel. (L2, PDF p. 382)
63. b Improved precision of test equipment is the listed subject that deals with the equipment used to test the aircraft. (L2, PDF p. 384)
64. a Approval is per aircraft and is applied for by the operator. (L2, PDF p. 384)
65. c The manual says each aircraft must be modified per the required service bulletins and is subject to the altitude monitoring programme. (L2, PDF p. 384)
66. b The training list names the part numbers of the air data computer. (L2, PDF p. 384)
67. b Where the instruments are electrical and the calculators separate, the mechanical conversion happens in those calculators and not in the instrument. (L3, PDF p. 376)
68. a The PFD is the screen that holds all the air data information, and it shows every one of the Basic T parameters. (L3, PDF p. 378)
69. b With QFE the barometric altimeter reads zero on the airfield, while with QNH it reads the elevation there. (L3, PDF p. 380)
70. a The transition altitude is the height at which the altimeter is changed from QNH to the standard setting. (L3, PDF p. 380)
71. b The transition level is the flight level at which the altimeter is changed from the standard setting back to QNH. (L3, PDF p. 380)
72. b Height is normally taken from the low range radio altimeter, and the barometric altimeter on QFE gives it only as long as the air pressure does not change. (L3, PDF p. 380)
73. c The common 1013.2 hPa setting in cruise is what ensures safe vertical separation, so an aircraft left on QNH no longer has it. (L3, PDF p. 380)
74. b Standard setting only applies above the transition altitude, so below it the altimeter stays on QNH. (L3, PDF p. 380)
75. a RVSM airspace is flight level airspace, and flight levels are flown with the standard pressure of 1013.2 hPa set. (L3, PDF p. 382)
76. b Above FL 290 the vertical separation is 1000 ft and that airspace is called RVSM airspace. (L3, PDF p. 382)
77. a Three of the four required items are fitted, but RVSM needs two independent altimeters and the aircraft has only one. (L3, PDF p. 382)
78. c One of the listed items of RVSM equipment is an autopilot able to maintain the altitude, and that is the function lost. (L3, PDF p. 382)
79. a Skin damage near the static source disturbs the static pressure and so the altitude reading. (L3, PDF p. 382)

80. c Losing RVSM capability means the aircraft must leave RVSM airspace quickly. (L3, PDF p. 384)

Airspeed

PDF pages 386–389 · 21 questions

LEVEL 1 — RECALL

81. What does the abbreviation IAS stand for ●
- a. Inertial airspeed
 - b. Indicated airspeed
 - c. Increased airspeed
82. Which airspeed is displayed in aircraft equipped with air data computers and glass cockpits ●
- a. The calibrated airspeed
 - b. The true airspeed
 - c. The equivalent airspeed
83. For which purpose is the true airspeed needed ●
- a. For engine trim purposes
 - b. For navigation purposes
 - c. For design data purposes

LEVEL 2 — UNDERSTAND AND EXPLAIN

84. An airspeed indicator with bellows is connected to the pitot and static systems. Which pressure does its indication directly follow? ●
- a. The static pressure as the difference between ambient and total pressure
 - b. The dynamic pressure as the difference between total and static pressure
 - c. The total pressure as the sum of dynamic and ambient pressure
85. Under which conditions does an airspeed indicator with bellows show the real speed of the aircraft ●
- a. At cruise altitude with standard conditions
 - b. At mean sea level with compressibility corrected
 - c. At mean sea level with standard conditions
86. An air data computer converts the indicated airspeed into the calibrated airspeed. Which two errors does it remove? ●
- a. The error from the indicator and the error from the static pressure source
 - b. The error from the pitot source and the error from air density
 - c. The error from the indicator and the error from air compressibility
87. A mechanical standby airspeed indicator with bellows is fed with total and static pressure. Which airspeed does it display? ●
- a. The equivalent airspeed
 - b. The indicated airspeed
 - c. The calibrated airspeed
88. Of the two speeds derived after the calibrated airspeed, which one is needed for navigation and which one is kept for design data ●
- a. the true airspeed and the indicated airspeed respectively
 - b. the equivalent airspeed and the true airspeed respectively
 - c. the true airspeed and the equivalent airspeed respectively

- 89.** Under which condition is the speed formula valid ●
- Only when the medium used is at standard temperature
 - Only when the medium used is above the tropopause
 - Only when the medium used is incompressible
- 90.** An aircraft accelerates to a high speed and the total pressure rises more than the speed formula predicts. What produces this additional pressure? ●
- Air expands at higher speeds and generates a total pressure decrease
 - Air compresses at lower speeds and generates a static pressure increase
 - Air compresses at higher speeds and generates a total pressure increase
- 91.** How does the compressibility error change with airspeed and altitude ●
- The higher the airspeed and altitude, the larger the error
 - The lower the airspeed and the higher the altitude, the larger the error
 - The higher the airspeed and altitude, the smaller the error
- 92.** What is the equivalent airspeed ●
- The calibrated airspeed after the compressibility error has been removed
 - The indicated airspeed after the instrument error has been removed
 - The true airspeed after the density calculation has been removed
- 93.** Why is the equivalent airspeed not shown to the crew of a commercial airplane ●
- It is only available when the air data computer is in test
 - It is only valid below the tropopause in standard conditions
 - It is only important for design data and further calculations
- 94.** A navigation computation needs the speed of the aircraft through the surrounding air mass. Which airspeed is this? ●
- The calibrated airspeed
 - The true airspeed
 - The equivalent airspeed
- 95.** In which order are the airspeeds derived from the pressure inputs ●
- CAS, IAS, TAS, EAS
 - IAS, EAS, CAS, TAS
 - IAS, CAS, EAS, TAS
- 96.** Which calculation is made between EAS and TAS
- A density calculation
 - A compressibility correction
 - A static source correction

LEVEL 3 — APPLY AND REASON

- 97.** A standby airspeed indicator with bellows and the airspeed readout of the glass cockpit are read at the same moment and differ. Which airspeeds are the two showing? ●
- Calibrated airspeed on the bellows instrument and indicated airspeed on the glass cockpit
 - Indicated airspeed on the bellows instrument and true airspeed on the glass cockpit
 - Indicated airspeed on the bellows instrument and calibrated airspeed on the glass cockpit

98. The indicated airspeed follows the dynamic pressure directly, and the speed formula behind it holds only for an incompressible medium. At high speed, which behaviour of the air affects the indication ●
- cools and generates a static pressure increase
 - expands and generates a total pressure decrease
 - compresses and generates a total pressure increase
99. Navigation needs the real speed of the aircraft measured against the surrounding air, but the cockpit shows only the airspeed indication. Which speed is needed and which value must be applied to obtain it? ●
- The true airspeed, obtained by applying static pressure as a corrective value
 - The equivalent airspeed, obtained by applying temperature as a corrective value
 - The true airspeed, obtained by applying temperature as a corrective value
100. Why is the equivalent airspeed lower than the calibrated airspeed at high speed and altitude? ●
- Because compression of the air lowers the static pressure, so the indication is too low
 - Because reduced density lowers the total pressure, so the indication is too high
 - Because compression of the air raises the total pressure, so the indication is too high
101. Two aircraft fly at the same calibrated airspeed, one at 20,000 ft and one at 40,000 ft. Which one has the larger compressibility error?
- The aircraft at 40,000 ft
 - Both have the same error
 - The aircraft at 20,000 ft

Answers — Airspeed

81. **b** The text defines IAS as indicated airspeed. (L1, PDF p. 386)
82. **a** Glass cockpits with an ADC display CAS. (L1, PDF p. 386)
83. **b** The manual states TAS is required for navigation. (L1, PDF p. 388)
84. **b** IAS follows dynamic pressure, which is total pressure minus static pressure. (L2, PDF p. 386)
85. **c** The bellows instrument is only true at mean sea level in standard conditions. (L2, PDF p. 386)
86. **a** CAS is IAS corrected for instrument error and static source error. (L2, PDF p. 386)
87. **b** A bellows instrument with no computer correction shows IAS. (L2, PDF p. 386)
88. **c** The true airspeed is the one wanted for navigation; the equivalent airspeed is not indicated and serves design data and further calculations. (L2, PDF p. 388)
89. **c** The formula assumes an incompressible medium. (L2, PDF p. 388)
90. **c** Compression of the air at high speed raises the total pressure, which the instrument reads as extra speed. (L2, PDF p. 388)
91. **a** The error grows with both airspeed and altitude. (L2, PDF p. 388)
92. **a** EAS is what is left when the compressibility error is taken out of CAS. (L2, PDF p. 388)
93. **c** EAS is a design and calculation value, not a cockpit indication. (L2, PDF p. 388)
94. **b** TAS is the aircraft speed relative to the surrounding air mass. (L2, PDF p. 388)
95. **c** The summary chain runs IAS to CAS to EAS to TAS. (L2, PDF p. 389)
96. **a** The last step of the chain is the density calculation. (L2, PDF p. 389)
97. **c** A bellows instrument shows the indicated airspeed, while the glass cockpit indication is the calibrated airspeed. (L3, PDF p. 386)
98. **c** IAS depends directly on the dynamic pressure, and at higher speeds the air compresses and adds to the total pressure, which the formula does not allow for. (L3, PDF p. 388)

- 99. c** Navigation needs the true airspeed, and temperature is the corrective value applied so that the true airspeed can be indicated. (L3, PDF p. 388)
- 100. c** The extra total pressure from compression makes the reading too high, so the correction reduces the speed. (L3, PDF p. 388)
- 101. a** The error grows with altitude, so the higher aircraft has the larger error. (L3, PDF p. 389)

Barometers, capsules and instrument errors

PDF pages 390–398 · 31 questions

LEVEL 1 — RECALL

102. What forms the sensing element of an aneroid barometer ●
- a. A small chamber made of thin glass
 - b. A small capsule made of thin metal
 - c. A small tube filled with thin fluid
103. What is the function of a Bourdon tube ●
- a. It converts pressure into a proportional electrical signal
 - b. It converts pressure into a proportional temperature signal
 - c. It converts vibration into a proportional electrical signal
104. Which cross-sectional shape does the curved tube of a Bourdon transducer have ●
- a. An oval shape
 - b. A round shape
 - c. A square shape
105. Which stroke do diaphragm capsules supply as the measurable value ●
- a. 14 to 15 mm
 - b. 0.4 to 0.5 mm
 - c. 4 to 5 mm

LEVEL 2 — UNDERSTAND AND EXPLAIN

106. The aneroid barometer takes its name from a nerios, not fluid. What is its sensing element ●
- a. a curved tube with an oval cross section
 - b. a vertical tube filled with a fluid
 - c. a small capsule made of thin metal
107. What creates the negative pressure at the top of a liquid barometer tube
- a. The fluid flowing out of the tube due to its own weight
 - b. The expansion of the glass tube with temperature
 - c. The ambient air pressure acting on the container
108. How does an aneroid capsule react to increasing air pressure ●
- a. It is compressed
 - b. It is unchanged
 - c. It is expanded
109. Which pressure is applied to the outside of the capsule in an aneroid altimeter ●
- a. The static pressure
 - b. The dynamic pressure
 - c. The total pressure
110. Which additional provision, beyond the pressure capsule and the indicating linkage, is built into an aneroid altimeter
- a. A compensation for the effect of acceleration
 - b. A compensation for the effect of temperature
 - c. A compensation for the effect of humidity

- 111.** A Bourdon tube transducer is opened for inspection. Which two parts carry out the pressure conversion? ●
- The straight tube and a position sensor
 - The curved tube and a drive coil
 - The curved tube and a position sensor
- 112.** A Bourdon tube transducer is fitted with a synchro transmitter. Which quantity does this sensor convert into an electrical signal? ●
- The pressure inside the tube
 - The temperature of the tube wall
 - The position of the tube tip
- 113.** Which effect do suitable bearings, clean construction of all moving parts and, in the altimeter, support with vibration guard against? ●
- hysteresis
 - backlash
 - friction
- 114.** Of the two capsule error influences, which one depends on whether the pressure was rising or falling? ●
- the elastic aftereffect
 - the hysteresis
 - the effect of temperature
- 115.** A diaphragm capsule delivers only a stroke of 4 to 5 mm with a small force. What must the transfer mechanism do with it? ●
- Convert it into a damped rotation of the needle
 - Convert it into an amplified stroke of the capsule
 - Convert it into an amplified rotation of the needle
- 116.** Which effect does friction have in the transfer mechanism of an aneroid instrument? ●
- It falsifies the measured result
 - It damps the capsule stroke
 - It raises the capsule hysteresis
- 117.** How is friction in the indicator mechanism kept to a minimum? ●
- With suitable bearings and clean construction of all moving parts
 - With sealed bearings and grease filling of all moving parts
 - With heavier levers and increased spring tension on all moving parts
- 118.** In the altimeter one further measure helps to reduce the effect of friction. Which measure is this? ●
- Vibration of the instrument
 - Balancing of the needle
 - Damping of the capsule
- 119.** Play in the gearing between the pinion and the tooth segment would falsify the indication. How is this play taken out? ●
- By increased bearing clearance
 - By spring tension on both sides
 - By spring tension on one side

- 120.** What is meant by the hysteresis of a membrane capsule ●
- The stroke lags behind the pressure when a constant pressure acts for a long time
 - The curve with increasing pressure differs from the curve with decreasing pressure
 - The curve with increasing pressure differs from the curve with increasing temperature
- 121.** What causes the elastic aftereffect of a capsule ●
- A rapidly changing pressure acting on the capsule during a climb
 - A temperature change acting on the capsule mounting
 - A constant pressure acting on the capsule over a longer period of time
- 122.** What is the result of the slow yielding of a capsule under constant pressure ●
- A time lag in the stroke
 - A permanent offset of the needle
 - A change of the residual pressure
- 123.** Friction in the transfer mechanism, hysteresis, elastic aftereffect and temperature all work against precise operation. Which instruments do these error influences affect? ●
- All aneroid type instruments
 - Only metric scale altimeters
 - Only digital air data computers

LEVEL 3 — APPLY AND REASON

- 124.** Static pressure acting on the capsule of an aneroid altimeter produces a stroke of only a few millimetres with a small force. Which part makes this usable for the indication ●
- the pressure scale, which sets the reference level
 - the position sensor, which converts it into an electrical signal
 - the transfer mechanism, which amplifies it into rotation of the needle
- 125.** The synchro transmitter of a Bourdon tube transducer is found unserviceable while the tube itself is sound. Which part of the conversion is lost? ●
- The supply of static pressure to the curved tube
 - The deformation of the curved tube by the applied pressure
 - The conversion of the tube tip position into an electrical signal
- 126.** An altimeter needle must turn 51 times over its full range. Which demand does this place on the instrument?
- The highest demands on case sealing and static line length
 - The highest demands on material and fine mechanical precision
 - The highest demands on electrical power and signal bandwidth
- 127.** The spring that holds one side of the gearing under tension is found broken in an aneroid instrument. Which error influence is no longer compensated? ●
- Hysteresis of the membrane capsule
 - Backlash between the pinions and the tooth segments
 - Friction in the bearings of the moving parts
- 128.** Which measures are used to characterize the properties of a capsule for its intended use ●
- Special coatings, artificial ageing and particular bearing materials
 - Special alloys, natural ageing and particular residual pressures
 - Special alloys, artificial ageing and particular waveforms

- 129.** Which parts make temperature compensation demanding over the range -70 to +50 degrees C ●
- The capsule mounting and the flexible tubes in the static system
 - The capsule mounting and the variable steering levers in the indicator mechanism
 - The pinion bearings and the quick release connectors of the instrument
- 130.** An altimeter is held at one steady pressure on the test bench for a long time and the indication slowly drifts. Which error influence explains this? ●
- The backlash of the pinion gearing
 - The elastic aftereffect of the capsule
 - The hysteresis of the capsule
- 131.** An altimeter reads slightly differently at the same test pressure depending on whether the pressure was raised or lowered to reach it. Which error influence is shown? ●
- Friction in the transfer mechanism
 - Elastic aftereffect of the diaphragm capsule
 - Hysteresis of the diaphragm capsule
- 132.** Through which parts must the smallest pressure change act to deflect the needle exactly
- From the capsule via bellows, resonator and coil
 - From the capsule via levers, shaft and cogwheel
 - From the diaphragm via strain gauges, bridge and amplifier

Answers — Barometers, capsules and instrument errors

- 102. b** The aneroid element is a thin metal capsule. (L1, PDF p. 390)
- 103. a** The Bourdon tube arrangement turns pressure into a proportional electrical output. (L1, PDF p. 392)
- 104. a** The tube has an oval cross section in the unpressurised state. (L1, PDF p. 392)
- 105. c** The capsule stroke is only 4 to 5 mm. (L1, PDF p. 398)
- 106. c** An aneroid barometer senses pressure with a thin metal capsule and so needs no liquid at all. (L2, PDF p. 390)
- 107. a** The weight of the fluid draws it down and leaves a negative pressure above it. (L2, PDF p. 390)
- 108. a** Rising pressure squeezes the capsule, so it contracts. (L2, PDF p. 390)
- 109. a** The aircraft aneroid altimeter senses static pressure acting on an evacuated capsule. (L2, PDF p. 391)
- 110. b** The aneroid altimeter carries a temperature compensation in addition to the capsule and the indicating linkage. (L2, PDF p. 391)
- 111. c** Only the curved oval tube and a position sensor at its tip are required. (L2, PDF p. 392)
- 112. c** The sensor senses tip position, not pressure directly. (L2, PDF p. 392)
- 113. c** Those three measures are the ones named for keeping friction to a minimum. (L2, PDF p. 398)
- 114. b** Hysteresis is the difference between the rising and the falling pressure curve, while the elastic aftereffect comes from pressure held constant over a long time. (L2, PDF p. 398)
- 115. c** The mechanism amplifies the tiny stroke into needle rotation. (L2, PDF p. 398)
- 116. a** Friction in the mechanism corrupts the reading. (L2, PDF p. 398)
- 117. a** Good bearings and clean mechanical construction minimise friction. (L2, PDF p. 398)
- 118. a** Vibration keeps the altimeter mechanism from sticking, helping against friction. (L2, PDF p. 398)
- 119. c** A one-sided spring tension takes up the gear backlash. (L2, PDF p. 398)
- 120. b** Hysteresis is the difference between the rising and falling pressure characteristics. (L2, PDF p. 398)
- 121. c** Long-held constant pressure makes the capsule yield slowly. (L2, PDF p. 398)
- 122. a** The stroke lags in time, which is the elastic aftereffect. (L2, PDF p. 398)
- 123. a** The manual states these errors act on all aneroid instruments. (L2, PDF p. 398)

- 124. c** Static pressure moves the capsule only a few millimetres, so the transfer mechanism has to turn that small stroke into an amplified needle rotation. (L3, PDF p. 391)
- 125. c** The synchro transmitter is the position sensor, so with it unserviceable the tube tip still moves but its position is no longer turned into an electrical signal. (L3, PDF p. 392)
- 126. b** Such fine resolution needs the best material and fine mechanics. (L3, PDF p. 396)
- 127. b** The one-sided spring tension is what takes out the backlash in the gearing, so a broken spring leaves that backlash uncompensated. (L3, PDF p. 398)
- 128. c** The three listed measures are special alloys, artificial ageing and particular waveforms. (L3, PDF p. 398)
- 129. b** Compensation is hard because of the capsule mounting and the variable steering levers. (L3, PDF p. 398)
- 130. b** Slow drift under a constant held pressure is the elastic aftereffect, not hysteresis. (L3, PDF p. 398)
- 131. c** A difference between the rising and falling approach to the same pressure is hysteresis. (L3, PDF p. 398)
- 132. b** The mechanical chain runs capsule, levers, shaft, cogwheel, needle. (L3, PDF p. 398)

Pressure transducers

PDF pages 400–407 · 28 questions

LEVEL 1 — RECALL

- 133.** What is the function of a pressure transducer ●
- To convert electrical signals into measured pressures
 - To convert measured pressures into mechanical needle movement
 - To convert measured pressures into electrical signals
- 134.** In which equipment are pressure transducers used ●
- In air data computers
 - In flight control computers
 - In inertial reference units
- 135.** Several electronic pressure sensor principles exist. Which one is described as the most commonly used? ●
- The vibrating transducer
 - The force compensated cell
 - The piezoresistive transducer
- 136.** From which material is the micromechanical pressure transducer chamber made ●
- Silicon material
 - Metal material
 - Pyrex material
- 137.** What is the size of the micromechanical pressure transducer structure ●
- Only a few micrometres
 - Only a few centimetres
 - Only a few millimetres
- 138.** How many strain gauges are normally fitted on the diaphragm of a piezoresistive transducer ●
- Two resistors
 - Eight resistors
 - Four resistors

LEVEL 2 — UNDERSTAND AND EXPLAIN

- 139.** Pressure transducers are fitted in air data computers because the computer needs the measured pressures in which form ●
- as electrical signals
 - as an output frequency of the indicator
 - as amplified needle rotations
- 140.** Pressure transducers have converted the measured pressures into electrical signals. What happens with these signals next?
- They are sent to the computer and transformed into the required output signals
 - They are sent to the probe heater and transformed into the required current
 - They are sent to the indicator and transformed into the required needle deflection

- 141.** What does the output frequency generated by a vibrating transducer depend on ●
- the supply voltage
 - the input pressure
 - the reference chamber vacuum
- 142.** Which output does a vibrating transducer generate ●
- An output frequency which depends on the input pressure
 - An output voltage which depends on the input pressure
 - An output frequency which depends on the reference pressure
- 143.** Which function does the outer case of a vibrating transducer have
- It forms the outer wall of the measured pressure chamber
 - It forms the inner wall of the reference pressure chamber
 - It forms the outer wall of the reference pressure chamber
- 144.** Which type of barometer chamber is the micromechanical pressure transducer built like ●
- the liquid-based type
 - the mercury type
 - the aneroid type
- 145.** What happens when pressure acts on the silicon lid of a piezoresistive transducer
- The lid is dented and the resistance in its surface changes
 - The lid vibrates and the frequency of its surface changes
 - The lid is dented and the capacitance in its surface changes
- 146.** What is the output voltage of a piezoresistive transducer proportional to ●
- The reference pressure
 - The supply frequency
 - The input pressure
- 147.** The four strain gages on the diaphragm have to produce one measuring voltage. In which configuration are they connected? ●
- A wheatstone bridge
 - A follow-up linkage
 - A synchro transmitter
- 148.** How are the four strain gauges distributed on the base plate
- All four on the front side of the plate
 - Three on the front side and one on the back side
 - Two on the front side and two on the back side

LEVEL 3 — APPLY AND REASON

- 149.** A force compensated pressure cell is fitted instead of a plain pressure cell. Which two advantages are gained?
- It is sensitive to pressure changes and has small instrument errors
 - It is insensitive to pressure changes and has small instrument errors
 - It is insensitive to pressure changes and needs no separate pressure cells
- 150.** Which pressure sensing type is listed as needing no different pressure cells for static and pitot?
- The force compensated pressure cell
 - The vibration pressure transducer
 - The plain pressure cell type

- 151.** An electronic pressure sensor delivers an output frequency that depends on the input pressure. Which sensor is this ●
- the piezoresistive transducer, the most commonly used type
 - the vibrating transducer, the most commonly used type
 - the pressure cell transducer, the most commonly used type
- 152.** One transducer gives an output frequency depending on the input pressure, another an output voltage in proportion to the input pressure. Which two transducers are these ●
- the vibrating transducer and the micromechanical transducer respectively
 - the micromechanical transducer and the vibrating transducer respectively
 - the vibrating transducer and the Bourdon tube transducer respectively
- 153.** A vibrating transducer is cut open. Where is the pressure that is being measured found?
- In the reference chamber outside the vibrating cylinder
 - In the pressure chamber inside the center body
 - In the pressure chamber inside the vibrating cylinder
- 154.** What is done with the signal from the pickup coil
- It is amplified and used as output, and fed back to the reference chamber
 - It is digitized and used as reference, and fed back to the drive coil
 - It is amplified and used as output, and fed back to the drive coil
- 155.** Why is the pickup coil signal fed back to the drive coil of a vibrating transducer?
- To keep the reference chamber at a constant vacuum
 - To keep the output frequency at a constant value
 - To keep the cylinder vibrating with its natural frequency
- 156.** A pressure sensor consists of a chamber made of silicon, only a few millimetres in size. Which sensor is this, and how is its chamber built ●
- the micromechanical transducer, built as a liquid-based chamber
 - the vibrating transducer, built as an aneroid-type chamber
 - the micromechanical transducer, built as an aneroid-type chamber
- 157.** Pressure bends the diaphragm of a piezoresistive transducer and the resistance of the strain gages changes. What does the bridge then deliver, and what is it proportional to? ●
- An output frequency, in proportion to the input pressure
 - An output voltage, in proportion to the input pressure
 - An output voltage, in proportion to the supply voltage
- 158.** Pressure is applied to a piezoresistive transducer. How do the two bridge voltages behave?
- U_1 decreases while U_2 increases
 - U_1 increases while U_2 decreases
 - U_1 and U_2 both increase
- 159.** A strain gauge wire is made thicker while its length stays the same. How does its resistance change?
- It is independent of the area
 - It increases with increasing area
 - It decreases with increasing area
- 160.** A strain gauge on the bent side of the diaphragm is stretched. Why does its resistance rise?
- Because the material constant of the wire rises with the strain
 - Because the wire becomes shorter and its cross section larger
 - Because the wire becomes longer and its cross section smaller

Answers — Pressure transducers

- 133. c** Transducers turn pressure into electrical signals. (L1, PDF p. 400)
- 134. a** The manual names air data computers as an application. (L1, PDF p. 400)
- 135. a** The manual calls the vibrating transducer the most commonly used type. (L1, PDF p. 402)
- 136. a** The chamber is made of silicon. (L1, PDF p. 404)
- 137. c** The structure measures only a few millimetres. (L1, PDF p. 404)
- 138. c** Normally four strain gauges are used. (L1, PDF p. 406)
- 139. a** A transducer turns the measured pressures into electrical signals, which is the form the air data computer works with. (L2, PDF p. 400)
- 140. a** The signals go to the computer, which makes the required outputs. (L2, PDF p. 400)
- 141. b** The frequency the vibrating transducer puts out is set by the pressure fed into it. (L2, PDF p. 402)
- 142. a** Its output is a frequency that varies with the measured pressure. (L2, PDF p. 402)
- 143. c** The outer case is the outer wall of the reference chamber. (L2, PDF p. 402)
- 144. c** Although it is made of silicon, the micromechanical transducer is described as an aneroid-type chamber. (L2, PDF p. 404)
- 145. a** Denting the lid deforms the surface resistors and changes their resistance. (L2, PDF p. 404)
- 146. c** The output voltage rises in proportion to the applied pressure. (L2, PDF p. 404)
- 147. a** The normally four strain gages of the diaphragm are connected in a wheatstone bridge configuration. (L2, PDF p. 406)
- 148. c** Two gauges sit on the front and two on the back. (L2, PDF p. 406)
- 149. b** The force compensated cell is insensitive to pressure changes and keeps instrument errors small. (L3, PDF p. 401)
- 150. b** The table gives 'no different pressure cells for static and pitot' for the vibration type; both pressure cell types are listed as needing different cells for static and pitot. (L3, PDF p. 401)
- 151. b** The vibrating transducer is named as the most commonly used type and it is the one whose output frequency depends on the input pressure. (L3, PDF p. 402)
- 152. a** The vibrating transducer produces a frequency; the micromechanical transducer produces a voltage in proportion to the pressure. (L3, PDF p. 402)
- 153. c** The measured pressure sits in the chamber inside the cylinder. (L3, PDF p. 402)
- 154. c** The pickup signal serves both as the amplified output and as the feedback to the drive coil. (L3, PDF p. 402)
- 155. c** The feedback loop sustains vibration at the natural frequency. (L3, PDF p. 402)
- 156. c** The micromechanical transducer is the aneroid-type chamber made of silicon whose structure is only a few millimetres in size. (L3, PDF p. 404)
- 157. b** The four strain gages sit in a wheatstone bridge, and the change in their resistance gives an output voltage in proportion to the input pressure. (L3, PDF p. 406)
- 158. b** The two halves of the bridge move in opposite directions when the plate bends. (L3, PDF p. 406)
- 159. c** A larger cross section lowers the resistance. (L3, PDF p. 406)
- 160. c** Figure 35 shows the stretched side getting longer and thinner, and $R = \rho \times l / A$, so both changes raise the resistance. (L3, PDF p. 407)

Pitot-static systems

PDF pages 408–426 · 86 questions

LEVEL 1 — RECALL

- 161.** Which pressure is used by an altimeter ●
- a. Static pressure only
 - b. Pitot and static pressure
 - c. Pitot pressure only
- 162.** An airspeed indicator in a simple system is connected to two pressure lines. Which pressures does it use? ●
- a. Static pressure and differential pressure
 - b. Pitot pressure and total pressure
 - c. Pitot pressure and static pressure
- 163.** How many separate pitot-static pressure systems are installed on modern commercial aircraft ●
- a. Three or four separate systems
 - b. One or two separate systems
 - c. Two or three separate systems
- 164.** What is static pressure ●
- a. The actual atmospheric pressure surrounding the aircraft
 - b. The build-up of air pressure in a tube facing the airstream
 - c. The difference between total pressure and ram pressure
- 165.** A pneumatic device is calibrated to the international standard atmosphere. Which value of air density at mean sea level applies? ●
- a. 1,225 kg/m³
 - b. 1,013 kg/m³
 - c. 1,150 kg/m³
- 166.** What does the abbreviation SSE stand for ●
- a. Static source error
 - b. Standard source error
 - c. Static system error
- 167.** What is pitot pressure ●
- a. The build-up of air pressure in a tube with its opening facing the airstream
 - b. The build-up of air pressure in a tube with its opening across the airstream
 - c. The actual atmospheric pressure in the air surrounding the aircraft
- 168.** Which two pressures make up the total pressure P_{tot} ●
- a. Ram pressure and differential pressure
 - b. Static pressure and differential pressure
 - c. Ram pressure and static pressure
- 169.** Where is the pitot tube usually located on the aircraft ●
- a. Near the nose, on the nose itself or below the cockpit
 - b. Near the tail, on the fin itself or below the rudder
 - c. Near the nose, on the wing leading edge or below the flaps

- 170.** What does the pitot tube consist of and which pressure is present in it during the entire flight ●
- A flapper plate chamber in which total pressure P_{tot} is present
 - A cylindrical pipe in which total pressure P_{tot} is present
 - A cylindrical pipe in which static pressure P_s is present
- 171.** Why is the Prandtl tube electrically heated ●
- To prevent condensation
 - To prevent deformation
 - To prevent icing
- 172.** A Prandtl tube supplies both pressures from one probe. Where is it located on the aircraft? ●
- Under the wing of the aircraft
 - At the rear of the aircraft
 - At the front of the aircraft
- 173.** An air data system receives three parameters from one single multi function probe. Which parameters are they? ●
- Angle of sideslip, total air temperature and total pressure
 - Angle of attack, total air temperature and total pressure
 - Angle of attack, static air temperature and static pressure
- 174.** Which part of the multi function probe measures the angle of attack ●
- A wind vane
 - A TAT tube
 - A pitot probe

LEVEL 2 — UNDERSTAND AND EXPLAIN

- 175.** What do the altimeter and the vertical speed indicator have in common ●
- Both use pitot pressure
 - Both use pitot pressure and static pressure
 - Both use static pressure
- 176.** Why are the left and right hand static pressure measuring locations of one static system connected to each other ●
- To increase the pressure delivered to the connected instruments
 - To drain any water collected inside the static pressure lines
 - To compensate for any unbalance in pressure measurement
- 177.** Compared with the simple system of a small aircraft, what does a modern commercial aircraft have ●
- One pitot system feeding two static systems
 - One pitot system and one static system as well
 - Two or three separate pitot-static pressure systems
- 178.** Why are modern commercial aircraft fitted with separate pitot-static pressure systems ●
- For weight reasons
 - For calibration reasons
 - For safety reasons
- 179.** What does the alternate static source selector valve allow the Captain to do ●
- Use the copilot's static pressure system
 - Use the copilot's pitot pressure system
 - Isolate the copilot's static pressure system

- 180.** The Captain's static system becomes faulty in flight. Which statement about switching over to the auxiliary system by means of the static air source selector valve is correct?
- Only on a few aircraft types can the indicators be switched to the pitot system
 - Only on a few aircraft types can the indicators be switched to the auxiliary system
 - On every aircraft type the indicators can be switched to the auxiliary system
- 181.** Between what does the handle of the static source selector valve choose for the instruments ●
- Normal static pressure and alternate static pressure
 - Normal static pressure and pitot pressure
 - Alternate static pressure and pitot pressure
- 182.** Why are static ports made flush with the skin of the aircraft ●
- The surface stays smooth and no pressure distorting turbulences occur
 - Water entering the ports is drained away again
 - The ports can be heated electrically from inside the skin
- 183.** On which two quantities does air pressure p depend ●
- Air density and mach number
 - Air density and temperature
 - Temperature and angle of attack
- 184.** What is done with the ISA values defined at mean sea level
- All pneumatic devices are corrected according to the flap position
 - All pneumatic devices are calibrated according to these values
 - All pneumatic devices are heated according to these values
- 185.** What do aircraft manufacturers ensure when they choose the position of the static ports ●
- That they lie in areas with high turbulence and ram air distortion
 - That they lie in areas with low turbulence and ram air distortion
 - That they lie in areas with low temperature and air density variation
- 186.** A technician is told that the static port zones must be absolutely clean and have a smooth surface. Which reason does the warning give? ●
- So that no pressure distorting turbulences can occur
 - So that no electrical heating current can be lost
 - So that no water can enter the connected pressure lines
- 187.** Why do most helicopters have special cover plates at the static ports ●
- To compensate the effects caused by the strong vibration of the rotor
 - To compensate the effects caused by the high angle of attack of the rotor
 - To compensate the effects caused by the strong downwash of the rotor
- 188.** Among the factors the static source error depends on, which one is qualified as applying only where relevant ●
- The angle of attack
 - The position of the flaps
 - The mach number
- 189.** Why can a static source error arise on an aircraft ●
- It is not always possible to calibrate the pneumatic devices to the ISA values
 - It is not always possible to find turbulence and ram air free zones on the aircraft
 - It is not always possible to keep the static ports electrically heated in flight

- 190.** An engineer is told that the static source error of a jet aeroplane is not a constant value. On which factors does it depend? ●
- The mach number, the angle of attack, the rudder position and the nose gear position
 - The mach number, the angle of attack, the flap position and the nose gear position
 - The mach number, the air density, the flap position and the outside air temperature
- 191.** How large is the static source error on slow-flying propeller planes ●
- So large that it must be corrected
 - So small that it must be measured
 - So small that it is negligible
- 192.** The crew has to obtain the correct value from the distorted values being displayed. Where are the correction tables found?
- In the Operating Manual or in the flight regulations
 - In the Operating Manual or in the weight and balance data
 - In the Aircraft Maintenance Manual or in the flight log
- 193.** Which pressure reading is affected by a static source error ●
- The total pressure reading
 - The ram pressure reading
 - The static pressure reading
- 194.** The static source error is listed with several dependencies. Which of them is marked as unusual? ●
- The angle of attack
 - The position of the nose gear
 - The position of the flaps
- 195.** Why does the static pressure have to be known before the ram pressure can be obtained ●
- Static pressure and ram pressure are measured by the same probe
 - Total pressure is made up of ram pressure plus static pressure
 - Ram pressure is the atmospheric pressure surrounding the aircraft
- 196.** What is the difference between total pressure and ram pressure ●
- Total pressure also contains the static pressure
 - Total pressure is measured at the static ports
 - Total pressure also contains the dynamic pressure
- 197.** Which pressure alone is required to measure the speed of the aircraft in relation to the ambient air ●
- The static pressure p_s
 - The ram pressure q
 - The total pressure P_{tot}
- 198.** How is ram pressure q obtained in the instrument ●
- By measuring the sum of the total pressure and the static pressure
 - By measuring the differential pressure between total and static pressure
 - By measuring the static pressure at two separate static ports
- 199.** Ram pressure cannot be sensed on its own. Which pressure is already known from the static ports and is used to obtain it? ●
- The static pressure p_s
 - The total pressure P_{tot}
 - The dynamic pressure q

- 200.** How does the electrical heating of the pitot tube compare with that of the static ports ●
- The pitot tube is heated, while only some aircraft types have heated static ports
 - Only the static ports are heated, the pitot tube being self draining
 - Both are heated on every aircraft type
- 201.** Why is a pitot tube aligned towards the predetermined airstream ●
- The airstream keeps the drain hole clear of water
 - The airstream stops the tube from freezing up
 - Pitot pressure builds up in a tube whose opening faces the airstream
- 202.** A pitot tube is being refitted to an aeroplane after removal. How must it be aligned? ●
- So that it faces the predetermined rotor downwash
 - So that it lies across the predetermined airstream
 - So that it faces the predetermined airstream
- 203.** What do the small drain hole and the baffle of a pitot tube ensure ●
- That ram air entering the tube is slowed down
 - That water entering the tube is drained away
 - That the tube inlet is kept free of any deformation
- 204.** A student is worried that the drain hole in a pitot tube lets pressure escape. Which statement from the manual answers this? ●
- The drain hole and the baffle slightly reduce the total pressure
 - The drain hole and the baffle do not affect the total pressure
 - The drain hole and the baffle slightly reduce the static pressure
- 205.** During a walk-around a technician finds a small dent at the inlet of a pitot tube. Which instruction from the manual applies?
- The tube inlet must not display any signs of deformation whatsoever
 - The tube inlet may display slight deformation if the drain hole is clear
 - The tube inlet may display slight deformation if the heating is serviceable
- 206.** Why is the Prandtl tube called a pitot-static-tube ●
- It is fitted at the front of the aircraft like a pitot tube
 - It combines a pitot tube with a sensor to measure static pressure
 - It is heated electrically like the static ports
- 207.** How does the fitting position of the Prandtl tube compare with that of the pitot tube ●
- The Prandtl tube is located in the same way as the pitot tube at the front of the aircraft
 - The Prandtl tube is located on the sides of the fuselage and the pitot tube at the front
 - The Prandtl tube is located below the rotor and the pitot tube at the front
- 208.** On a Prandtl tube, what do the cylindrical inlet and the holes drilled in the sides supply, in that order ●
- The total pressure and the connected static lines
 - The total pressure and the drain line
 - The connected static lines and the total pressure
- 209.** What does the Prandtl tube combine in one unit ●
- A pitot tube with a sensor to measure static pressure
 - A pitot tube with a sensor to measure total air temperature
 - A static port with a sensor to measure the angle of attack

- 210.** A Prandtl tube has a cylindrical inlet at the front and a row of holes drilled in its sides. What do these side holes supply? ●
- Drainage for one or two connected water traps
 - Information for one or two connected static lines
 - Information for one or two connected pitot lines
- 211.** What do the flapper plate and drain hole of a Prandtl tube prevent? ●
- Water from entering the pitot line
 - Icing from forming at the tube inlet
 - Roughness from forming on the tube surface
- 212.** A Prandtl tube is found with a rough surface after a repair. Which consequence does the manual warn about?
- It will lead to total pressure errors
 - It will lead to static pressure errors
 - It will lead to heating element failures
- 213.** How does a Prandtl tube obtain the ram pressure and static pressure needed for speed indication? ●
- From one single ram air tube
 - From two connected static ports
 - From two separate measuring points
- 214.** Which two separate measuring points does the Prandtl tube replace? ●
- The pitot tube and the air data computer
 - The pitot tube and the static port
 - The static port and the difference bellows
- 215.** A Prandtl tube delivers total pressure and static pressure. Where is the required differential pressure obtained? ●
- In the static port or in the drain hole
 - In the tube itself or in the piping
 - In the instrument or in a computer
- 216.** A multi function probe must deliver the total pressure to the air data system. Which of its parts provides this parameter? ●
- The temperature tube
 - The pitot probe
 - The wind vane
- 217.** Each multi function probe has to pass its measured data to the related computer. Through which item does it do so?
- Through the digital converter
 - Through the air data computer
 - Through the difference bellows
- 218.** How are the two pneumatic back up instruments arranged in some aircraft types? ●
- Both functions are integrated in one instrument
 - Only the altitude function is retained
 - Both functions are shared between three instruments

LEVEL 3 — APPLY AND REASON

- 219.** The pitot probe of a simple system becomes completely blocked while the static system stays clear. Which of the three supplied instruments is affected? ●
- The altimeter
 - The vertical speed indicator
 - The airspeed indicator
- 220.** Which instrument of the simple system needs a pressure connection in addition to the static system ●
- The vertical speed indicator
 - The altimeter
 - The airspeed indicator
- 221.** During a repair the connection between the left and right hand measuring locations of one static pressure system is left open. What is lost by this? ●
- The supply of pitot pressure to the instruments
 - The compensation for any unbalance in pressure measurement
 - The heating of the static measuring locations
- 222.** A technician traces one static pressure system of a small aeroplane and finds a measuring location on the left and on the right hand side of the fuselage. Which statement describes this arrangement correctly? ●
- Each location belongs to a separate static system and they stay isolated
 - Both locations belong to the same static system and are connected together
 - Both locations belong to the same pitot system and are connected together
- 223.** The Captain moves the handle of the alternate static source selector valve to its other position. Which static pressure then reaches his instruments? ●
- The copilot's static pressure system
 - His own static pressure system
 - The pitot pressure of his own system
- 224.** An aeroplane has a fault in one pilot's static system and is not one of the few types that can be switched to the auxiliary system. Which possibility remains? ●
- Use of the other pilot's static system via the selector valve
 - Use of the pitot system via the selector valve
 - Use of the auxiliary system via the selector valve
- 225.** A helicopter cannot keep its static ports in air that is free of rotor downwash. Which fitting deals with this? ●
- Special cover plates at the static ports
 - A drain hole at each static port
 - Electrically heated static ports
- 226.** After a skin repair the zone around a static port is left with a rough surface. What must be expected at that port? ●
- A blockage of the pressure pipe
 - A loss of the port heating
 - Pressure distorting turbulences

- 227.** Why are static ports placed in areas with low turbulence and ram air distortion ●
- Because turbulence and ram air distortion overheat the static port zones
 - Because turbulence and ram air distortion falsify the static pressure reading
 - Because turbulence and ram air distortion block the static pressure hoses
- 228.** An aeroplane extends its flaps on the approach. Which reading can change as a result of this alone? ●
- The static pressure reading
 - The total air temperature reading
 - The angle of attack reading
- 229.** Of the factors the static source error depends on, which two change with the configuration of the aircraft ●
- The mach number and the position of the flaps
 - The position of the flaps and the position of the nose gear
 - The mach number and the angle of attack
- 230.** An aeroplane changes its angle of attack in cruise with everything else unchanged. Which measurement may be affected by this? ●
- The total pressure measurement
 - The pitot pressure measurement
 - The static pressure measurement
- 231.** A slow-flying propeller plane and a jet aeroplane near the speed of sound differ in one of the listed static source error factors above all. Which factor is that? ●
- The position of the flaps
 - The position of the nose gear
 - The mach number
- 232.** A jet aeroplane flies near to the speed of sound and its static source error exceeds the tolerances required by law. Which correction becomes necessary?
- A correction using an alternate static source valve
 - A correction using an air data computer
 - A correction using a differential pressure bellows
- 233.** A static source error falsifies the static pressure supplied to a simple pitot-static system. Which instruments are affected? ●
- All three instruments of the simple system
 - Only the vertical speed indicator of the system
 - Only the airspeed indicator of the system
- 234.** A slow-flying propeller aeroplane and a jet aeroplane near the speed of sound both have a static source error. Which comparison is correct? ●
- Negligible on the jet aeroplane and possibly beyond the legal tolerance on the propeller plane
 - Beyond the legal tolerance on both aeroplanes and corrected by the air data computer
 - Negligible on the propeller aeroplane and possibly beyond the legal tolerance on the jet
- 235.** In the relation $P_{tot} = q + p_s$, where is the quantity p_s delivered to the instrument from ●
- The static ports
 - The pitot probe
 - The drain hole of the pitot tube

- 236.** Why is an airspeed indicator fed from the pitot probe as well as the static system ●
- The static pressure alone gives the speed in the surrounding air
 - The pitot probe cannot deliver a pressure without a static reference
 - The ram pressure it needs is the total pressure less the static pressure
- 237.** A pitot tube delivers the pressure present in its cylindrical pipe throughout the flight. What must be supplied in addition before a speed can be obtained? ●
- The total air temperature from a temperature probe
 - The static pressure from the static ports
 - A second total pressure from another pitot tube
- 238.** An aircraft type is to take both the ram pressure and the static pressure needed for speed indication from one single ram air tube. Which probe is used? ●
- A multi function probe
 - A pitot tube
 - A Prandtl tube
- 239.** Which feature does a Prandtl tube carry that a plain pitot tube does not ●
- Holes drilled in the sides supplying connected static lines
 - A cylindrical inlet for the total pressure
 - Electrical heating of the tube
- 240.** Both a pitot tube and a Prandtl tube keep water out of the pitot line. Which item does this in the Prandtl tube together with the drain hole? ●
- A flapper plate
 - A cover plate
 - A baffle
- 241.** On an aircraft type fitted with a Prandtl tube, where does the static pressure needed for speed indication come from ●
- The same single ram air tube that supplies the ram pressure
 - The cylindrical inlet at the front of the tube
 - A pair of static ports on the sides of the fuselage
- 242.** A Prandtl tube delivers total pressure and static pressure to a computer. Which quantity does the computer obtain from them, and what is it needed for? ●
- The ram pressure, needed for the speed indication
 - The static source error, needed for the altitude indication
 - The total air temperature, needed for the speed indication
- 243.** An air data system is to receive the angle of attack from the same probe that gives it the total pressure. Which probe meets this? ●
- A Prandtl tube
 - A multi function probe
 - A pitot tube
- 244.** A multi function probe delivers the total pressure to the air data system. Which quantity is still required for a speed indication ●
- The static pressure
 - The angle of attack
 - The total air temperature

- 245.** An aircraft with a modern EFIS carries two pneumatic back up instruments. Which of them has to be connected to the pitot system? ●
- a. The standby altimeter
 - b. Both of them equally
 - c. The standby airspeed indicator
- 246.** With what must the standby altimeter fitted as a back up in an EFIS aircraft be supplied ●
- a. Pitot pressure only
 - b. Pitot pressure and static pressure
 - c. Static pressure only

Answers — Pitot-static systems

- 161. a** The manual states the altimeter works on static pressure alone. (L1, PDF p. 408)
- 162. c** The airspeed indicator is the only one of the three that needs both pitot and static pressure. (L1, PDF p. 408)
- 163. c** The manual gives two or three separate systems. (L1, PDF p. 410)
- 164. a** Static pressure is simply the ambient atmospheric pressure around the aircraft. (L1, PDF p. 412)
- 165. a** The ISA sea level density given in the manual is 1,225 kg/m³. (L1, PDF p. 412)
- 166. a** SSE is the static source error. (L1, PDF p. 414)
- 167. a** Pitot pressure builds up in a tube whose opening points into the airstream. (L1, PDF p. 416)
- 168. c** The manual gives total pressure as ram pressure plus static pressure. (L1, PDF p. 416)
- 169. a** These are the three locations the manual names. (L1, PDF p. 418)
- 170. b** It is a cylindrical pipe carrying total pressure throughout the flight. (L1, PDF p. 418)
- 171. c** Electrical heating keeps the probe free of ice. (L1, PDF p. 420)
- 172. c** It sits at the front, like the pitot tube. (L1, PDF p. 420)
- 173. b** AOA, TAT and total pressure are the three outputs of the probe. (L1, PDF p. 422)
- 174. a** The wind vane is the AOA sensing element. (L1, PDF p. 422)
- 175. c** Static pressure is named as the pressure used by each of these two instruments. (L2, PDF p. 408)
- 176. c** Cross-connecting the two sides evens out an unbalanced pressure reading between them. (L2, PDF p. 408)
- 177. c** The simple system has one of each, whereas the modern commercial aircraft has two or three separate systems. (L2, PDF p. 410)
- 178. c** Redundant systems are installed for safety. (L2, PDF p. 410)
- 179. a** The valve switches the Captain over to the copilot's static source. (L2, PDF p. 410)
- 180. b** The manual restricts this switching possibility to a few aircraft types only. (L2, PDF p. 410)
- 181. a** The valve is connected to a normal and an alternate static pressure supply and passes one of them to the instruments. (L2, PDF p. 411)
- 182. a** A flush port keeps the zone smooth, which is what the warning requires so that no pressure distorting turbulences arise. (L2, PDF p. 412)
- 183. b** The manual lists air density and temperature as the two quantities. (L2, PDF p. 412)
- 184. b** The ISA values are the calibration reference for pneumatic devices. (L2, PDF p. 412)
- 185. b** Low turbulence and low ram air distortion keep the static reading true. (L2, PDF p. 412)
- 186. a** Dirt or roughness creates turbulence that distorts the sensed pressure. (L2, PDF p. 412)
- 187. c** The cover plates offset the pressure effects of the rotor downwash. (L2, PDF p. 412)
- 188. a** The angle of attack is the item carrying the qualification if applicable. (L2, PDF p. 414)
- 189. b** Where no turbulence free and ram air free zone exists, the sensed static pressure is falsified. (L2, PDF p. 414)
- 190. b** These four items are the dependencies the manual lists for the SSE. (L2, PDF p. 414)
- 191. c** At low speed the error is negligible. (L2, PDF p. 414)

- 192. a** The manual names the Operating Manual and the flight regulations as the sources of the correction tables. (L2, PDF p. 414)
- 193. c** The SSE falsifies the static pressure reading. (L2, PDF p. 414)
- 194. b** Only the nose gear position carries the remark unusual. (L2, PDF p. 414)
- 195. b** Since total pressure is the sum of ram and static pressure, the static part must be known so that it can be taken out. (L2, PDF p. 416)
- 196. a** Total pressure is ram pressure together with static pressure, so the static part is what it holds in addition. (L2, PDF p. 416)
- 197. b** Only the dynamic or ram pressure q is needed for speed measurement. (L2, PDF p. 416)
- 198. b** Ram pressure is the difference between total and static pressure. (L2, PDF p. 416)
- 199. a** The static ports supply p_s , which is subtracted from P_{tot} to leave q . (L2, PDF p. 416)
- 200. a** Heating is stated plainly for the pitot tube, whereas heated static ports are stated only for some aircraft types. (L2, PDF p. 418)
- 201. c** Pitot pressure only builds up when the opening of the tube faces the airstream, so the alignment is what produces the pressure. (L2, PDF p. 418)
- 202. c** The tube must point into the airstream to build up total pressure. (L2, PDF p. 418)
- 203. b** Their job is to get rid of water that has entered the tube. (L2, PDF p. 418)
- 204. b** The manual states in brackets that they do not affect the total pressure. (L2, PDF p. 418)
- 205. a** The manual allows no deformation of the inlet at all. (L2, PDF p. 418)
- 206. b** The name follows from the tube holding both a pitot tube and a static pressure sensor in one unit. (L2, PDF p. 420)
- 207. a** The two probes are fitted the same way, at the front of the aircraft. (L2, PDF p. 420)
- 208. a** Total pressure enters at the cylindrical inlet while the side holes feed the static lines. (L2, PDF p. 420)
- 209. a** It puts the pitot function and a static pressure sensor into one probe. (L2, PDF p. 420)
- 210. b** The side holes feed the static lines, while the front inlet feeds total pressure. (L2, PDF p. 420)
- 211. a** They keep water out of the pitot line. (L2, PDF p. 420)
- 212. b** Roughness disturbs the flow over the side holes and falsifies static pressure. (L2, PDF p. 420)
- 213. a** The Prandtl tube takes both pressures at one single tube. (L2, PDF p. 420)
- 214. b** The manual names the pitot tube and the static port as the two points it replaces. (L2, PDF p. 420)
- 215. c** The subtraction is done in the instrument or in a computer, not in the probe. (L2, PDF p. 420)
- 216. b** The built-in pitot probe supplies the total pressure. (L2, PDF p. 422)
- 217. a** The digital converter is the link between the probe and the related computer. (L2, PDF p. 422)
- 218. a** The two standby functions can be brought together in a single instrument on some types. (L2, PDF p. 426)
- 219. c** Of the three instruments only the airspeed indicator uses pitot pressure, so only it depends on the pitot probe. (L3, PDF p. 408)
- 220. c** The altimeter and vertical speed indicator use static pressure only, while the airspeed indicator also needs pitot pressure. (L3, PDF p. 408)
- 221. b** The two locations are joined only to compensate unbalance in pressure measurement, so breaking the joint removes that compensation. (L3, PDF p. 408)
- 222. b** One static system uses both side locations and they are joined to each other. (L3, PDF p. 408)
- 223. a** The valve exists so that the Captain can take his static pressure from the copilot's system instead of his own. (L3, PDF p. 410)
- 224. a** Switching to the auxiliary system is limited to a few types, so the cross-over to the other pilot's static system is what remains. (L3, PDF p. 410)
- 225. a** The cover plates are fitted for exactly this purpose, to compensate the effects of the strong rotor downwash at the ports. (L3, PDF p. 412)
- 226. c** The zone is required to be smooth precisely so that no pressure distorting turbulences occur, so a rough surface allows them. (L3, PDF p. 412)

- 227. b** Where such zones cannot be found the static pressure reading is affected, so a quiet zone keeps the reading true. (L3, PDF p. 412)
- 228. a** The static source error depends on the position of the flaps, and it is the static pressure reading that this error affects. (L3, PDF p. 414)
- 229. b** Flaps and nose gear are the two listed items that are set by the aircraft configuration, the other two being flight condition values. (L3, PDF p. 414)
- 230. c** The angle of attack is one of the factors the static source error depends on, and that error affects the static pressure reading. (L3, PDF p. 414)
- 231. c** The two cases are set apart by how near the speed of sound they fly, which is the mach number among the listed factors. (L3, PDF p. 414)
- 232. b** The manual calls for an air data computer correction in this case. (L3, PDF p. 414)
- 233. a** All three instruments of the simple system are supplied with static pressure, so all three are affected. (L3, PDF p. 414)
- 234. c** The manual calls the error negligible at low speed and possibly beyond the legal tolerance near the speed of sound. (L3, PDF p. 414)
- 235. a** The static pressure p_s used in the relation is the pressure measured at the static ports. (L3, PDF p. 416)
- 236. c** Speed needs the ram pressure, and the ram pressure is obtained by subtracting the static pressure from the total pressure delivered by the pitot probe. (L3, PDF p. 416)
- 237. b** The pitot tube gives total pressure only, and the static pressure from the static ports has to be subtracted from it to get the ram pressure. (L3, PDF p. 418)
- 238. c** The Prandtl tube is the single ram air tube that supplies both of these pressures. (L3, PDF p. 420)
- 239. a** Both probes have a cylindrical pipe and electrical heating, but the side holes feeding static lines belong to the Prandtl tube. (L3, PDF p. 420)
- 240. a** The Prandtl tube works with a flapper plate and a drain hole, the baffle being the item named for the pitot tube. (L3, PDF p. 420)
- 241. a** On these types the two pressures come from the one Prandtl tube instead of separate pitot tube and static port. (L3, PDF p. 420)
- 242. a** The differential of the two delivered pressures is the ram pressure q , which is what the speed indication requires. (L3, PDF p. 420)
- 243. b** Only the multi function probe is stated to give angle of attack as well as total pressure. (L3, PDF p. 422)
- 244. a** The ram pressure needed for speed is the total pressure less the static pressure, and the probe supplies only the total pressure of the two. (L3, PDF p. 422)
- 245. c** An airspeed indicator uses pitot pressure as well as static pressure, while an altimeter uses static pressure only. (L3, PDF p. 426)
- 246. c** It is a pneumatic altimeter, and an altimeter works on static pressure alone. (L3, PDF p. 426)

Pitot-static leaks and blockages

PDF pages 428–448 · 63 questions

LEVEL 1 — RECALL

247. Which colour is used to designate pitot system connections ●
- Yellow
 - Red
 - Blue
248. Two lines are parted at a quick-disconnect during maintenance. What happens at the coupling? ●
- It vents the line to ambient air
 - It seals itself
 - It stays open until a cap is fitted
249. A technician must check the mouth of a static port against its limits. Where are these limits normally found? ●
- In ATA chapter 21
 - In ATA chapter 31
 - In ATA chapter 34
250. A technician must assess damage to the skin next to a static port. Which equipment is used? ●
- A scale and a feeler-gage
 - A straight edge and a torque wrench
 - A scale and a depth micrometer
251. Which system controls the cabin rate ●
- The anti-ice system
 - The pitot static system
 - The airconditioning system
252. Which instrument is affected by a leakage in the pitot system ●
- The vertical speed indicator only
 - The airspeed indicator only
 - The altimeter only
253. What do the altimeter and the vertical speed indicator indicate during a climb with a blocked static system ●
- The altimeter increases and the vertical speed indication stays at zero
 - The altimeter holds the same altitude and the vertical speed indication stays at zero
 - The altimeter holds the same altitude and the vertical speed indication rises
254. What does the vertical speed indicator indicate during a descent with a blocked static system ●
- A steady rate of descent
 - A zero rate
 - A steady rate of climb
255. What does a completely blocked pitot tube mean ●
- The inlet and the drain hole are blocked
 - The inlet is blocked and the drain hole is open
 - The drain hole is blocked and the inlet is open

- 256.** A pitot tube is completely blocked by ice. Which instruments are not affected? ●
- The airspeed indicator and the vertical speed indicator
 - The altimeter and the airspeed indicator
 - The altimeter and the vertical speed indicator
- 257.** The pitot inlet is blocked by ice but the water drain hole is still open. What does the airspeed indication do? ●
- It increases into the overspeed region
 - It decreases to zero
 - It stays unchanged at the last value
- 258.** The indications fluctuate during a check of the pitot static system. What does this show? ●
- The presence of a leak in them
 - The presence of water in them
 - The presence of ice at the static port

LEVEL 2 — UNDERSTAND AND EXPLAIN

- 259.** An instrument has been changed in the pitot static system. What must be carried out? ●
- An approved flushing of the static system
 - An approved accuracy check of the static ports
 - An approved leak-proof test
- 260.** What is the purpose of the scale and the feeler-gage used at a static port ●
- To assess damage of the aircraft skin against the limits
 - To check that the static port zone is smooth enough to prevent turbulences
 - To measure the mouth of the static port against the limits
- 261.** Why must the static port zones have a smooth surface ●
- So that no turbulences can occur
 - So that water can drain away from the port
 - So that no ice can form on the port
- 262.** Where are the limits for damage of the aircraft skin at the static ports found ●
- In the ATA chapter for air conditioning or the wiring manual
 - In the ATA chapter for the aircraft structure or the SRM
 - In the ATA chapter for the instruments or the IPC
- 263.** What do the results of a leakage depend on
- The size of the leakage and the altitude of the aircraft
 - The position of the leakage and the outside air temperature
 - The size of the leakage and whether the cabin is pressurized
- 264.** A static line leaks inside the unpressurized cabin of an aircraft in flight. What does the altimeter do? ●
- It slightly increases the indicated altitude
 - It slightly decreases the indicated altitude
 - It freezes at the altitude flown when the leak started
- 265.** What effect does a static system leakage inside an unpressurized cabin have on the airspeed indicator ●
- It shows zero indicated airspeed
 - It shows a slightly lower indicated airspeed
 - It shows a slightly higher indicated airspeed

- 266.** Why do the altimeter and the airspeed indicator decrease their indication when a static line leaks into a pressurized cabin ●
- The pressure in the static lines increases
 - The pressure in the static lines decreases
 - The pitot and the static pressure become nearly the same
- 267.** What effect does a static system leakage into a pressurized cabin have on the pressure in the static lines ●
- The pressure decreases
 - The pressure increases
 - The pressure stays unchanged
- 268.** What do the altimeter and the airspeed indicator indicate after a static leakage into a pressurized cabin ●
- Both stay unchanged
 - Both decrease
 - Both increase
- 269.** A pitot line leaks inside an unpressurized cabin. What does the airspeed indication do? ●
- It stays at the value flown when the leak started
 - It decreases to zero
 - It increases into the overspeed region
- 270.** Why does the airspeed indication decrease when both the pitot and the static system leak ●
- Pitot and static pressure tend to be the same
 - The total pressure in the pitot line is trapped at a constant value
 - The static pressure in the lines rises above the pitot pressure
- 271.** What effect does a leakage in both the pitot and the static system have on the airspeed indication ●
- It is not affected
 - It decreases
 - It increases
- 272.** Both the pitot system and the static system are leaking. What do the altimeter and the vertical speed indicator show? ●
- The same results as for a leakage in the pitot system
 - The same results as for a leakage in the static system
 - Correct indications, because the two leaks cancel each other
- 273.** What are the indications with a blocked static system while the aircraft flies at constant altitude and constant airspeed ●
- The altitude indication slowly increases
 - The airspeed indication slowly decreases
 - There are no changes in the indication
- 274.** The static system is blocked and the aircraft climbs. What does the airspeed indicator show? ●
- Zero airspeed
 - A decreasing airspeed indication
 - An increasing airspeed indication

- 275.** What do the altimeter and the vertical speed indicator indicate with a blocked static system, during a climb and during a descent ●
- The same altitude and a vertical speed that follows the aircraft
 - The same altitude and a vertical speed of zero
 - A changing altitude and a vertical speed of zero
- 276.** The static system is blocked and the aircraft descends. What does the airspeed indicator show? ●
- A decreasing airspeed indication
 - An increasing airspeed indication
 - Zero airspeed
- 277.** Why does the airspeed indicator react like an altimeter when the pitot tube is totally blocked ●
- Total and static pressure become nearly equal, so the indication falls away
 - The static pressure is trapped, so only the changing total pressure moves the indication
 - The total pressure is trapped, so only the changing static pressure moves the indication
- 278.** What does the airspeed indicator do during a climb with a totally blocked pitot tube ●
- It stays at the speed flown when the blockage occurred
 - It increases and can reach the overspeed region
 - It decreases and can reach zero
- 279.** How does the airspeed indicator behave when the pitot tube is totally blocked ●
- It reacts like a vertical speed indicator
 - It reacts like an altimeter
 - It reacts like a turn coordinator
- 280.** What does the airspeed indicator do during a descent with a totally blocked pitot tube ●
- It remains at the value held during cruise
 - It increases to a higher indicated airspeed
 - It decreases to a lower indicated airspeed
- 281.** What happens to the pitot pressure when the pitot tube inlet is blocked but the water drain hole is still open ●
- It stays at the value it had before the blockage
 - It decreases to the value of the static pressure
 - It increases to the value of the total pressure
- 282.** Why are the altimeter and the vertical speed indicator not affected by a blocked pitot inlet ●
- They get both pitot and static pressure
 - They only get pitot pressure
 - They only get static pressure
- 283.** Why can water be present in the pressure pipes ●
- They are not drained on a regular basis
 - They are capped for protection during testing
 - Their test fittings are used to simulate altitude and speed
- 284.** The static ports of a cold aeroplane are capped and the aircraft is moved into a warm hangar. What happens in the static system?
- The pressure stays the same
 - The pressure decreases
 - The pressure increases

- 285.** What happens to the ram pressure when both the pitot inlet and the drain are iced over ●
- It is always zero
 - It is constant at every altitude
 - It depends on the altitude

LEVEL 3 — APPLY AND REASON

- 286.** A device is replaced in lines with screw connections whose connectors are designated red. Which system is concerned and what action is required afterwards ●
- The static system, and an approved leak-proof test must be carried out
 - The pitot system, and the self-sealing connections make a test unnecessary
 - The pitot system, and an approved leak-proof test must be carried out
- 287.** A line was parted only at a quick-disconnect. What applies to the leak-proof test?
- It is required, as after any other parting
 - It is required, but only on the static side
 - It is not required in this case
- 288.** A technician must check the mouth of a static port and also damage to the skin around it. Where are the two sets of limits found? ●
- The port mouth limits in the structure chapter or the SRM and the skin damage limits in ATA chapter 34
 - Both the port mouth limits and the skin damage limits in ATA chapter 34
 - The port mouth limits in ATA chapter 34 and the skin damage limits in the structure chapter or the SRM
- 289.** A static line breaks inside an unpressurized cabin during flight. How does the vertical speed indicator behave? ●
- It jumps once and then shows correct data
 - It jumps once and then stays at zero
 - It slowly drifts into a permanent climb indication
- 290.** The same static line leak is considered in an unpressurized aircraft and in a pressurized one. How does the altimeter behave in each case? ●
- It slightly increases the indication in both aircraft
 - It slightly increases the indication in the unpressurized aircraft and decreases it in the pressurized one
 - It decreases the indication in the unpressurized aircraft and slightly increases it in the pressurized one
- 291.** How does the vertical speed indicator behave after a static system leakage in an unpressurized cabin compared with a pressurized cabin ●
- A small jump and then correct data in the unpressurized cabin, and an indication depending on cabin rate in the pressurized cabin
 - An indication depending on cabin rate in the unpressurized cabin, and a small jump and then correct data in the pressurized cabin
 - A small jump and then correct data in both cabins
- 292.** A static line leaks into a pressurized cabin. What does the vertical speed indication now depend on? ●
- The cabin rate
 - The aircraft rate of climb
 - The size of the leak

- 293.** A pitot line is leaking inside the cabin. How does the airspeed indication behave in an unpressurized aircraft compared with a pressurized one? ●
- It decreases to zero in the unpressurized aircraft and is unpredictable in the pressurized one
 - It decreases to zero in both aircraft
 - It is unpredictable in the unpressurized aircraft and decreases to zero in the pressurized one
- 294.** Why does the airspeed indication fall to zero with a pitot leakage in an unpressurized cabin ●
- The pitot pressure rises with the cabin pressure
 - Static and pitot pressure become nearly the same
 - The static pressure becomes much greater than the pitot pressure
- 295.** A pitot line leaks inside a pressurized cabin. What can be said about the airspeed indication? ●
- It always falls to zero, because pitot and static pressure become equal
 - It is unpredictable, because it depends on altitude and cabin pressure
 - It always rises, because the cabin pressure is higher than the static pressure
- 296.** Which instruments are disturbed by a leakage in the pitot system alone compared with a leakage in both the pitot and the static system ●
- The airspeed indicator and the altimeter, compared with the airspeed indicator and the vertical speed indication
 - The airspeed indicator and the vertical speed indication, compared with the airspeed indicator alone
 - The airspeed indicator alone, compared with the airspeed indicator, the altimeter and the vertical speed indication
- 297.** Both the pitot system and the static system leak inside an unpressurized cabin. What does the altimeter do? ●
- It slightly increases the indicated altitude
 - It holds the altitude it had when the leakage started
 - It slightly decreases the indicated altitude
- 298.** An aircraft cruises at constant altitude and constant airspeed with a blocked static system and then begins a climb. What does the altimeter do? ●
- It continues to display the same altitude
 - It jumps once and then follows the climb correctly
 - It begins to increase its indication as the climb starts
- 299.** Why does the airspeed indication decrease during a climb with a blocked static system ●
- The total pressure from the pitot tube decreases while the trapped static pressure stays constant
 - The total pressure and the static pressure both decrease together
 - The total pressure from the pitot tube stays constant while the static pressure decreases
- 300.** How does the airspeed indication with a blocked static system during a climb compare with a descent ●
- Decreasing during both the climb and the descent
 - Decreasing during the climb and increasing during the descent
 - Increasing during the climb and decreasing during the descent
- 301.** Why does the airspeed indication increase during a descent with a blocked static system ●
- The total pressure and the static pressure both increase together
 - The total pressure from the pitot tube increases while the trapped static pressure remains constant
 - The trapped static pressure increases while the total pressure remains constant

- 302.** Two aircraft climb, one with a blocked static system and one with a completely blocked pitot tube. What does the airspeed indicator show in each? ●
- A decreasing indication with the blocked static system and an increasing indication with the blocked pitot tube
 - An increasing indication with the blocked static system and a decreasing indication with the blocked pitot tube
 - A decreasing indication in both aircraft
- 303.** How does the airspeed indication during a descent with a blocked static system compare with a completely blocked pitot tube ●
- Decreasing with the blocked static system and increasing with the blocked pitot tube
 - Increasing with both blockages
 - Increasing with the blocked static system and decreasing with the blocked pitot tube
- 304.** The pitot tube becomes totally blocked in cruise and the thrust is then increased to accelerate. What does the airspeed indicator show? ●
- A rise in the indicated speed
 - No change in the indicated speed
 - A fall of the indicated speed to zero
- 305.** The pitot inlet of an aircraft is found blocked by ice. Which indication shows that the water drain hole is still open? ●
- The airspeed indication stops responding to changes of thrust
 - The airspeed indication decreases to zero
 - The airspeed indication increases into the overspeed region
- 306.** What happens to the pitot pressure and the airspeed indication when the pitot inlet is iced over but the drain is free ●
- The pitot pressure falls to static pressure and the airspeed indication increases
 - The pitot pressure remains constant and the airspeed indication decreases to zero
 - The pitot pressure falls to static pressure and the airspeed indication decreases to zero
- 307.** Two pitot probes are examined, one iced at the inlet only and one iced at the inlet and the drain. What does each condition say about the heating? ●
- Insufficient heating for the inlet only, and defective or deactivated heating when both are iced
 - Defective or deactivated heating for the inlet only, and insufficient heating when both are iced
 - Insufficient heating in both conditions
- 308.** The pitot inlet is iced over but the drain is free. What does this tell you about the heating? ●
- The heating is insufficient
 - The heating is working correctly
 - The heating is defective or deactivated
- 309.** Both the pitot inlet and the drain are iced over. What does this say about the heating and the pitot pressure? ●
- The heating is defective or deactivated and the pitot pressure falls to static pressure
 - The heating is defective or deactivated and the pitot pressure remains constant
 - The heating is insufficient and the pitot pressure falls to static pressure

Answers — Pitot-static leaks and blockages

- 247. b** Pitot connections carry the red designation; yellow belongs to the static side. (L1, PDF p. 428)
- 248. b** A self-sealing coupling closes the line itself the moment it is parted. (L1, PDF p. 428)

249. c The manual points to ATA chapter 34 for the static port mouth limits. (L1, PDF p. 430)
250. a The manual names the scale and the feeler-gage as the assessment tools. (L1, PDF p. 430)
251. c The manual names the airconditioning system as the controller of cabin rate. (L1, PDF p. 434)
252. b Only the airspeed indicator is fed from the pitot system. (L1, PDF p. 436)
253. b With static pressure trapped, neither instrument can register the climb. (L1, PDF p. 440)
254. b The trapped static pressure never changes, so the VSI sees no rate at all. (L1, PDF p. 442)
255. a Complete blockage covers both the inlet and the drain hole. (L1, PDF p. 444)
256. c Those two run on static pressure, which the pitot blockage does not touch. (L1, PDF p. 444)
257. b With pitot pressure equal to static pressure the airspeed indicator reads zero. (L1, PDF p. 446)
258. b Fluctuating indications are the sign of water lying in the pipes. (L1, PDF p. 448)
259. c Changing an instrument or a device calls for an approved leak-proof test. (L2, PDF p. 428)
260. a The scale and feeler-gage are the tools named for assessing skin damage against the AMM limits. (L2, PDF p. 430)
261. a A smooth zone keeps the airflow undisturbed so the port senses true static pressure. (L2, PDF p. 430)
262. b Skin damage is a structural item, so the structure chapter or the SRM holds those limits. (L2, PDF p. 430)
263. c Size of the leak and the pressurization state of the cabin are the two factors given. (L2, PDF p. 432)
264. a The lower cabin pressure reaching the static line reads as a slightly greater altitude. (L2, PDF p. 432)
265. c The airspeed indicator reads slightly high, in the same sense as the altimeter. (L2, PDF p. 432)
266. a Cabin pressure entering the static lines raises the pressure there, and both instruments read lower as a result. (L2, PDF p. 434)
267. b Cabin pressure is above the ambient static pressure, so the static lines see a rise. (L2, PDF p. 434)
268. b The raised static pressure drives both indications down. (L2, PDF p. 434)
269. b With the pitot line open to the cabin the airspeed indication falls away to zero. (L2, PDF p. 436)
270. a Both lines fill with the same cabin pressure, so the difference the indicator works on shrinks. (L2, PDF p. 438)
271. b With both lines open to the same cabin pressure the difference shrinks and the indication falls. (L2, PDF p. 438)
272. b Those two instruments see only static pressure, so the static leak result still applies. (L2, PDF p. 438)
273. c Nothing changes while the trapped static pressure still matches the real one. (L2, PDF p. 440)
274. b In the climb the falling total pressure works against a trapped static pressure, so the reading drops. (L2, PDF p. 440)
275. b PDF page 440 prints the frozen altitude and zero vertical speed for the climb and this page the same pair for the descent, since both instruments work from the trapped static pressure. (L2, PDF p. 442)
276. b In the descent the rising total pressure works against a trapped static pressure, so the reading climbs. (L2, PDF p. 442)
277. c With the total pressure held fixed the indication is driven only by static pressure, which is what an altimeter responds to. (L2, PDF p. 444)
278. b With trapped total pressure and falling static pressure the indication runs up. (L2, PDF p. 444)
279. b Its reading now follows altitude change instead of speed. (L2, PDF p. 444)
280. c The rising static pressure against a trapped total pressure drives the reading down. (L2, PDF p. 444)
281. b The open drain lets the trapped pressure bleed down to static. (L2, PDF p. 446)
282. c Both instruments are fed from the static system alone. (L2, PDF p. 446)
283. a The pipes are not drained routinely, so water can collect in them. (L2, PDF p. 448)
284. c Warming the trapped air in a sealed static system raises its pressure. (L2, PDF p. 448)
285. c With the total pressure held fixed, only the static pressure varies, so ram pressure follows altitude. (L2, PDF p. 448)
286. c Red designates the pitot connections, and changing an instrument or a device calls for an approved leak-proof test. (L3, PDF p. 428)
287. c A parting at a quick-disconnect is the one exception to the leak-proof test requirement. (L3, PDF p. 428)
288. c The mouth of the port is covered by ATA chapter 34, while skin damage is covered by the structure chapter or the SRM. (L3, PDF p. 430)
289. a The VSI reacts only to the sudden change and then works normally again. (L3, PDF p. 432)

- 290. b** On PDF page 432 the leak into an unpressurized cabin lowers the pressure in the lines, so the altimeter reads slightly higher; on this page the cabin pressure raises the line pressure, so it reads lower. (L3, PDF p. 434)
- 291. a** On PDF page 432 the unpressurized case gives a single small jump and then correct data; on this page the pressurized cabin makes the indication follow the cabin rate. (L3, PDF p. 434)
- 292. a** With cabin air in the static line the VSI follows the cabin rate, not the aircraft. (L3, PDF p. 434)
- 293. a** With no cabin pressure the two pressures equalise and the indication falls to zero, while cabin pressure makes the result depend on altitude and cabin pressure. (L3, PDF p. 436)
- 294. b** The airspeed indicator works on the difference between the two pressures, which is now almost nil. (L3, PDF p. 436)
- 295. b** With a pressurized cabin the two varying pressures make the indication unpredictable. (L3, PDF p. 436)
- 296. c** PDF page 436 prints that a pitot leak affects the airspeed indicator only, and this page adds the altimeter and the vertical speed indication once the static system leaks as well. (L3, PDF p. 438)
- 297. a** This page refers the altimeter back to the static leakage result, and PDF page 432 prints that result for an unpressurized cabin as a slightly increased indicated altitude. (L3, PDF p. 438)
- 298. a** The trapped static pressure cannot change, so the altimeter holds the same altitude in level flight and through the climb. (L3, PDF p. 440)
- 299. a** Only the total pressure falls; the blocked static side cannot follow it. (L3, PDF p. 440)
- 300. b** PDF page 440 prints the decreasing indication in the climb and this page the increasing one in the descent, because the trapped static pressure stays fixed while the total pressure falls and then rises. (L3, PDF p. 442)
- 301. b** Only the total pressure rises; the blocked static side stays where it was. (L3, PDF p. 442)
- 302. a** PDF page 440 prints the decreasing indication with the static side blocked, while this page prints the increasing one with the pitot side blocked, because the falling static pressure then drives the instrument. (L3, PDF p. 444)
- 303. c** PDF page 442 prints the increasing indication with the static side blocked, while this page prints the decreasing one with the pitot side blocked, because the rising static pressure then drives the instrument. (L3, PDF p. 444)
- 304. b** At constant altitude neither pressure changes, so the reading is frozen whatever the thrust. (L3, PDF p. 444)
- 305. b** With the drain open the pitot pressure falls to static and the indication drops to zero, which a completely blocked tube does not do. (L3, PDF p. 446)
- 306. c** The open drain lets the pitot pressure fall to static pressure, leaving no difference for the indicator to show. (L3, PDF p. 448)
- 307. a** Ice at the inlet alone points to heating that is too weak, while ice at both openings points to heating that is defective or switched off. (L3, PDF p. 448)
- 308. a** Ice only at the inlet points to heating that is working but not enough. (L3, PDF p. 448)
- 309. b** Complete icing means the heating has failed or is off, and the sealed pitot pressure cannot change. (L3, PDF p. 448)

Altimeters

PDF pages 454–466 · 39 questions

LEVEL 1 — RECALL

- 310.** What is measured in the barometric method of determining altitude ●
- a. Satellite signal delay
 - b. Radio wave delay
 - c. Air pressure
- 311.** A barometric altimeter is connected to one pneumatic line only. Which pressure does it measure? ●
- a. Pitot pressure
 - b. Total pressure
 - c. Static pressure
- 312.** What is the shape of the ISA pressure graph at low altitudes ●
- a. Strongly exponential
 - b. A series of steps
 - c. Nearly linear
- 313.** Which pressure decrease corresponds to an altitude increase of 30 ft at low altitude ●
- a. 10 hPa
 - b. 1 hPa
 - c. 30 hPa
- 314.** What does the baro set knob give the altimeter ●
- a. A new reference pressure
 - b. A new static pressure source
 - c. A new scale calibration in metres
- 315.** The crew is preparing for departure. Which pressure do they set on the altimeters? ●
- a. The actual pressure at ground level
 - b. The actual pressure related to sea level
 - c. The standard pressure of the ISA
- 316.** What name is given to the reference pressure used for QNE ●
- a. Standard
 - b. Ambient
 - c. Nautical
- 317.** Who may select the barometric correction on an altimeter, besides the pilots ●
- a. The air data computer
 - b. Air traffic control
 - c. Maintenance personnel

LEVEL 2 — UNDERSTAND AND EXPLAIN

- 318.** How does a barometer determine altitude ●
- a. By using the increase in pressure with increasing altitude
 - b. By using the delay of a reflected radio wave
 - c. By using the decrease in pressure with increasing altitude

- 319.** A method of determining altitude uses the WGS84 gravitation model as its reference instead of the ground. Which method is it?
- Barometer
 - GPS
 - Radio altimeter
- 320.** How is the interior of a barometric altimeter casing connected ●
- To the outside air via the static system
 - To a sealed reference vacuum
 - To the outside air via the pitot system
- 321.** An aircraft climbs. What happens inside its barometric altimeter? ●
- Pressure decreases, the bellows contract and the pointer rises
 - Pressure increases, the bellows expand and the pointer falls
 - Pressure decreases, the bellows expand and the pointer rises
- 322.** Which construction feature distinguishes an encoding altimeter ●
- An additional static port and a drum
 - Additional bellows and electrical connectors
 - Additional pointers and a second baro scale
- 323.** An encoding altimeter puts out an ALT signal. Which system receives it? ●
- The air data computer
 - The ATC transponder system
 - The static system
- 324.** Why is an encoding altimeter connected to the ATC transponder system? ●
- So that the transponder can set the baro correction of the altimeter
 - So that ground control can be informed about the altitude of the aircraft
 - So that the crew can be informed about the ground pressure
- 325.** In which aircraft are encoding altimeters found ●
- Some small aircraft
 - All aircraft fitted with an air data computer
 - Only aircraft without a transponder
- 326.** Why is an encoding altimeter fitted with electrical connectors ●
- To provide the ALT signal for the ATC transponder system
 - To drive the additional bellows inside the instrument
 - To receive the reference pressure from an air data computer
- 327.** How does the baro set knob work in a pneumatic indicator ●
- It moves the bellows stack alone
 - It directly turns the whole altimeter mechanism
 - It sends a correction signal to the air data computer
- 328.** A new reference pressure is selected on an electrical altitude indicator. How is the correction made? ●
- Electrically inside the indicator or via an air data computer
 - Pneumatically, by an additional set of bellows
 - Mechanically, by turning the whole altimeter mechanism

- 329.** An airport lies well above sea level and the altimeter is set to QNH. What does the instrument show on the ground?
- The height of the aircraft above the runway
 - Zero feet
 - The altitude of the airport related to mean sea level
- 330.** The aircraft is descending and reaches the Transition Level. What do the pilots do with the altimeters?
- They adjust them to 1013 hPa
 - They adjust them to the pressure used at the departure airport
 - They adjust them to the actual pressure at the destination airport
- 331.** How is a new reference pressure applied in a pneumatic altimeter compared with an electrical indicator
- By turning the whole altimeter mechanism, compared with a correction made electrically inside the indicator or via an air data computer
 - By turning the whole altimeter mechanism, in the pneumatic indicator and in the electrical indicator alike
 - By a correction made electrically inside the indicator or via an air data computer, compared with turning the whole altimeter mechanism
- 332.** Which baro correction makes the altimeter read zero on the ground
- QNH
 - QNE
 - QFE
- 333.** To which level is the pressure used for QFE related
- Ground level
 - The standard atmosphere
 - Mean sea level
- 334.** An aircraft is in flight with QFE set on its altimeter. What does the indication represent?
- The height above mean sea level
 - The altitude in the standard atmosphere
 - The height above the ground
- 335.** Why do pilots of medium and large aircraft normally not use QFE?
- Their baro scales cannot be set that low
 - The QFE value is not transmitted by data link
 - They can use a radio altimeter system
- 336.** An aircraft in flight has the baro correction for QNH set. What does the altimeter read?
- The height above mean sea level
 - The height above the ground
 - The height above the transition altitude
- 337.** A technician selects QNE on an altimeter. Which reference is then used?
- The ambient pressure at mean sea level on the day
 - The ambient pressure at the airport
 - The standard atmosphere with 1013 hPa

- 338.** Where is the QNE setting normally used ●
- On the ground, not in flight
 - During takeoff and landing only
 - In flight, not on the ground
- 339.** What is the advantage of QNE ●
- All aircraft in cruise flight use the same reference pressure
 - The altimeter reads zero at touchdown
 - Every aircraft reads its own height above the ground
- 340.** What does an altimeter in flight indicate with the baro correction for QFE compared with QNH ●
- The height above the ground with both corrections
 - The height above the ground with QFE and the height above mean sea level with QNH
 - The height above mean sea level with QFE and the height above the ground with QNH
- 341.** An altimeter set to QFE indicates 0 ft at touchdown because of which reference ●
- Its reference is the standard atmosphere with 1013 hPa
 - Its reference is the actual pressure related on ground level
 - Its reference is the actual pressure related on mean sea level

LEVEL 3 — APPLY AND REASON

- 342.** An altimeter is set to standard pressure. Under which condition is its indicated altitude equal to the true altitude? ●
- When the pressure at sea level is really 1013 hPa
 - When the airport elevation is zero
 - When the aircraft is above the transition altitude
- 343.** A crew keeps 1013 hPa set on the altimeters for the whole cruise. Why does the indication not give the true altitude everywhere along the route? ●
- Because the indication is only correct where the pressure at sea level really is 1013 hPa
 - Because the pressure graph of the standard atmosphere is only linear at low altitudes
 - Because all aircraft in cruise flight select a different reference pressure
- 344.** Why do all aircraft in cruise use one common selected pressure instead of the local pressure? ●
- So that they all keep the same altitude separation in low and high pressure areas
 - So that ground control can read the elevation of the airports
 - So that each of them indicates its true height above the ground
- 345.** An altimeter is left calibrated to 1013 hPa while the real pressure at sea level below the aircraft is lower. What will the instrument indicate?
- The correct altitude
 - Too high an altitude
 - Too low an altitude
- 346.** An aircraft passes the Transition Altitude after takeoff and later the Transition Level before landing. Which setting is made at each point? ●
- 1013 hPa at both the Transition Altitude and the Transition Level
 - The actual pressure at the departure airport at the Transition Altitude and 1013 hPa at the Transition Level
 - 1013 hPa at the Transition Altitude and the actual pressure at the destination airport at the Transition Level

- 347.** Two altimeters at the same airport, which lies above sea level, read differently on the ground, one showing zero and the other the elevation of the airport. Which baro corrections are set? ●
- QFE on the one showing zero and QNE on the one showing the elevation
 - QNH on the one showing zero and QFE on the one showing the elevation
 - QFE on the one showing zero and QNH on the one showing the elevation
- 348.** The QNE and the QNH altimeter calibration both refer the altitude to mean sea level. What is the difference between them ●
- QNE uses the actual pressure related to ground level while QNH uses the standard atmosphere with 1013 hPa
 - QNE uses the actual pressure related to mean sea level while QNH uses the standard atmosphere with 1013 hPa
 - QNE uses the standard atmosphere with 1013 hPa while QNH uses the actual pressure related to mean sea level

Answers — Altimeters

- 310. c** The barometer method measures air pressure; the other two quantities belong to the radio altimeter and to GPS. (L1, PDF p. 454)
- 311. c** The barometric altimeter is an aneroid barometer working on static pressure only. (L1, PDF p. 454)
- 312. c** Close to the ground the pressure against altitude curve is nearly a straight line. (L1, PDF p. 462)
- 313. b** The printed relationship at low altitude is 30 ft per 1 hPa. (L1, PDF p. 462)
- 314. a** The knob only changes the reference pressure the indication is based on. (L1, PDF p. 464)
- 315. b** Before takeoff the reference is the actual sea level pressure, which is the QNH setting. (L1, PDF p. 464)
- 316. a** The 1013 hPa reference is called standard. (L1, PDF p. 466)
- 317. c** The manual names pilots and maintenance personnel as the ones who select the baro correction. (L1, PDF p. 466)
- 318. c** Pressure falls as altitude rises, and the barometer uses that relationship. (L2, PDF p. 454)
- 319. b** WGS84 is the reference of the GPS method; the barometer uses pressure and the radio altimeter uses the ground. (L2, PDF p. 454)
- 320. a** Outside air reaches the inside of the case through the static system. (L2, PDF p. 454)
- 321. c** Falling pressure around the capsule lets the bellows expand, which drives the pointer up the scale. (L2, PDF p. 454)
- 322. b** The encoding altimeter carries extra bellows and electrical connectors on top of the normal altimeter. (L2, PDF p. 460)
- 323. b** The encoding altimeter feeds the ALT signal to the ATC transponder. (L2, PDF p. 460)
- 324. b** The purpose of the ALT signal is to pass the aircraft altitude to ground control. (L2, PDF p. 460)
- 325. a** The encoding altimeter is described as a fit found in some small aircraft. (L2, PDF p. 460)
- 326. a** The electrical connectors carry the ALT signal that the ATC transponder system needs. (L2, PDF p. 460)
- 327. b** In a pneumatic instrument the setting is made mechanically by turning the whole mechanism. (L2, PDF p. 464)
- 328. a** Electrical indicators have no mechanism to turn, so the correction is done electrically. (L2, PDF p. 464)
- 329. c** With sea level as the reference, an airport above sea level indicates its own elevation. (L2, PDF p. 464)
- 330. c** At the Transition Level the standard setting is replaced by the destination setting. (L2, PDF p. 464)
- 331. a** The pneumatic instrument is re-referenced mechanically, the electrical one electrically inside itself or through an air data computer. (L2, PDF p. 464)
- 332. c** QFE uses the ambient pressure at ground level, so the indication on the ground is zero. (L2, PDF p. 466)
- 333. a** QFE is the ambient pressure at ground level, unlike QNH which is related to mean sea level. (L2, PDF p. 466)
- 334. c** With the ground as reference the instrument shows height above the ground. (L2, PDF p. 466)
- 335. c** A radio altimeter already gives the height above the ground, so the QFE setting is not needed. (L2, PDF p. 466)
- 336. a** QNH refers everything to mean sea level, in flight as well as on the ground. (L2, PDF p. 466)
- 337. c** QNE is always the fixed standard atmosphere value, not a value measured on the day. (L2, PDF p. 466)

- 338. c** QNE is the in-flight setting; on the ground the airport settings are used. (L2, PDF p. 466)
- 339. a** The whole point of QNE is a single common reference for all aircraft in the cruise. (L2, PDF p. 466)
- 340. b** QFE refers the indication to the ground and QNH to mean sea level. (L2, PDF p. 466)
- 341. b** QFE takes the pressure at ground level as its reference, so the indication is zero on the ground. (L2, PDF p. 466)
- 342. a** The conversion from static pressure to altitude only matches the true altitude when the real sea level pressure is the calibration value. (L3, PDF p. 462)
- 343. a** Indicated altitude only equals true altitude where the real sea level pressure is 1013 hPa, and PDF page 464 adds that the altimeters cannot be reset for every point of the route. (L3, PDF p. 462)
- 344. a** A common reference keeps the vertical separation between aircraft consistent whatever the local pressure is. (L3, PDF p. 464)
- 345. b** With the reference set higher than the real sea level pressure, the instrument over-reads. (L3, PDF p. 464)
- 346. c** The common standard pressure is taken at the Transition Altitude on the way up and given back for the local pressure at the Transition Level on the way down. (L3, PDF p. 464)
- 347. c** QFE is referenced to the ground and reads zero there, while QNH is referenced to mean sea level and shows the airport elevation. (L3, PDF p. 466)
- 348. c** Both refer the altitude to mean sea level, but QNE always takes the standard pressure while QNH takes the actual one. (L3, PDF p. 466)

Airspeed indicators

PDF pages 468–476 • 28 questions

LEVEL 1 — RECALL

349. Which type of measuring instrument is the airspeed indicator ●
- a. A differential pressure meter
 - b. An absolute pressure meter
 - c. A differential temperature meter
350. In which unit is the scale of the airspeed indicator calibrated ●
- a. Kilometres per hour
 - b. Knots
 - c. Nautical miles
351. What does the green band on the airspeed indicator of a small aircraft show ●
- a. Normal Operating Range
 - b. Full Flap Operating Range
 - c. Caution range
352. What does the white band on the airspeed indicator of a small aircraft show ●
- a. Caution range
 - b. Full Flap Operating Range
 - c. Normal Operating Range
353. What does the red mark on a helicopter airspeed indicator show ●
- a. The speed that must never be exceeded
 - b. The range of transition lift
 - c. The maximum speed allowed for autorotation
354. What does the red-white mark on a helicopter airspeed indicator show ●
- a. The maximum speed allowed for autorotation under all conditions
 - b. The speed that must never be exceeded
 - c. The range of transition lift
355. Where does the aircraft move at the speed called true airspeed ●
- a. Relative to the airspeed indicator scale
 - b. Through the surrounding air
 - c. Over the ground
356. What is the speed shown on the indicator called ●
- a. Indicated airspeed
 - b. Ground speed
 - c. True airspeed

LEVEL 2 — UNDERSTAND AND EXPLAIN

357. A technician traces the pitot line to the airspeed indicator capsule. Which pressure does that line deliver to the capsule interior? ●
- a. Total pressure from the pitot system
 - b. Static pressure from the static system
 - c. Dynamic pressure from the static system

- 358.** The capsule of an airspeed indicator sits inside a sealed casing. Which pressure acts inside that casing? ●
- Dynamic pressure from the pitot system
 - Total pressure from the pitot system
 - Static pressure from the static system
- 359.** In an airspeed indicator, what does the difference between total pressure and static pressure give ●
- Total pressure
 - Dynamic pressure
 - Static pressure
- 360.** A technician is asked what the indicated airspeed value is used for. Which answer is correct?
- As a value for altitude and for the maximum rate of climb
 - As a value for lift and for the maximum load on the flaps
 - As a value for thrust and for the maximum cabin pressure
- 361.** An aircraft is fitted with a primary flight display instead of a mechanical airspeed indicator. How is airspeed presented? ●
- As a speed limit marking on the PFD
 - As a speed bug on the PFD
 - As a speed band on the PFD
- 362.** How are the airspeed limits shown in small aircraft
- A digital readout beside the airspeed indicator
 - A movable pointer driven by the air data computer
 - Coloured ranges and lines on the airspeed indicator scale
- 363.** An aeroplane is being flown with the airspeed in the yellow band. Which condition applies? ●
- It may only be used in smooth air without hard manoeuvres
 - It may only be used with the flaps extended
 - It may be used in any air, including hard manoeuvres
- 364.** The slight increase of VMO with altitude results from taking account of which error ●
- The instrument error of the IAS
 - The compressibility error of the IAS
 - The static source error of the IAS
- 365.** A crew wants a minimum speed indication in the cockpit. On which aircraft is it available, and where is it shown? ●
- Only on aircraft with a glass cockpit, on the speed tape of the PFD
 - On all jet aircraft, as a coloured band on the airspeed indicator scale
 - Only on piston engine aircraft, as a red line on the airspeed indicator
- 366.** How does the yellow band on the airspeed indicator of a small aircraft differ from the green band ●
- It is the caution range, to be used only in smooth air without hard manoeuvres, while the green band is the full flap operating range
 - It is the caution range, to be used only in smooth air without hard manoeuvres, while the green band is the normal operating range
 - It is the normal operating range, while the green band is the caution range, to be used only in smooth air without hard manoeuvres

- 367.** A helicopter airspeed indicator carries two essential marked limits. What makes them necessary?
- The engine power limits of the helicopter
 - The aerodynamic properties of the helicopter
 - The maximum take-off weight of the helicopter
- 368.** An aircraft is flown at sea level in standard conditions and its IAS equals its TAS. Which further condition must be met? ●
- With negligible compressibility error
 - With negligible instrument error
 - With negligible static source error
- 369.** TAS is obtained from IAS by correcting for instrument error, static source error, dynamic pressure error and which further quantity
- Magnetic variation
 - Aircraft weight
 - Air density

LEVEL 3 — APPLY AND REASON

- 370.** A jet aeroplane climbs from low altitude to high altitude. Which limit governs its maximum speed at the higher altitude? ●
- The maximum allowed Mach number MMO
 - The maximum operating velocity VMO
 - The never exceed speed VNE
- 371.** Why can the speed limit of a jet aeroplane at low altitude be read directly as an indicated airspeed? ●
- Because the limit there depends on dynamic pressure, and the indicated airspeed indicator is a dynamic pressure indicator
 - Because the limit there depends on dynamic pressure, and the indicated airspeed indicator is a total pressure indicator
 - Because the limit there depends on the speed of sound, and the indicated airspeed indicator is a dynamic pressure indicator
- 372.** How do VMO and MMO change as a jet aeroplane climbs ●
- VMO decreases while MMO stays constant or increases slightly
 - VMO stays constant or increases slightly while MMO decreases
 - VMO and MMO both decrease
- 373.** A helicopter airspeed indicator carries a red mark and a red-white mark. Which limit does each of them give? ●
- The red mark the speed that must never be exceeded, the red-white mark the maximum speed allowed for autorotation
 - The red mark the speed that must never be exceeded, the red-white mark the range of transition lift
 - The red mark the maximum speed allowed for autorotation, the red-white mark the speed that must never be exceeded
- 374.** An aircraft climbs while holding a constant indicated airspeed. What happens to the difference between IAS and TAS? ●
- It decreases, because the air density decreases
 - It increases, because the air density decreases
 - It stays the same, because the IAS is corrected for density

- 375.** An aircraft is flown at sea level in standard conditions, but its static source error is not negligible. Do the indicated airspeed and the true airspeed coincide? ●
- No, because they coincide only when the static source error is negligible as well
 - Yes, because at sea level the two speeds always coincide
 - Yes, because the static source error only separates them at altitude
- 376.** An aircraft is not fitted with an air data computer. What must be provided instead? ●
- A second airspeed indicator for comparison
 - Correction tables in the Flight Operations Manual
 - Correction tables in the aircraft technical log

Answers — Airspeed indicators

- 349. a** The instrument works on the difference between two pressures, so it is a differential pressure meter. (L1, PDF p. 468)
- 350. b** The printed text gives knots as the unit of the scale. (L1, PDF p. 468)
- 351. a** Green marks the normal operating range on the scale. (L1, PDF p. 470)
- 352. b** White marks the range in which the flaps may be fully used. (L1, PDF p. 470)
- 353. a** Red marks VNE, the never exceed speed. (L1, PDF p. 472)
- 354. a** The red-white mark is the autorotation speed limit. (L1, PDF p. 472)
- 355. b** TAS is defined as the aircraft speed through the surrounding air mass. (L1, PDF p. 474)
- 356. a** What the instrument displays is the indicated airspeed. (L1, PDF p. 474)
- 357. a** The capsule interior receives total pressure, which the pitot system delivers. (L2, PDF p. 468)
- 358. c** The casing is fed from the static system, so static pressure surrounds the capsule. (L2, PDF p. 468)
- 359. b** Total pressure minus static pressure is by definition the dynamic pressure. (L2, PDF p. 468)
- 360. b** The manual lists lift and maximum load on the flaps and airframe among the uses of the IAS. (L2, PDF p. 468)
- 361. c** The third type of airspeed indication listed is the speed band shown on the PFD. (L2, PDF p. 468)
- 362. c** Small aircraft carry the limits as colour markings painted on the instrument scale. (L2, PDF p. 470)
- 363. a** The caution range is restricted to smooth air with no hard manoeuvres. (L2, PDF p. 470)
- 364. b** The manual attributes the slight rise to the compressibility error of the IAS. (L2, PDF p. 470)
- 365. a** The minimum speed indication is a glass cockpit feature shown on the PFD speed tape. (L2, PDF p. 470)
- 366. b** The green band is the normal operating range while the yellow band is a caution range with conditions attached to its use. (L2, PDF p. 470)
- 367. b** The manual gives the aerodynamic properties of the helicopter as the reason for the two marked limits. (L2, PDF p. 472)
- 368. c** The manual gives two conditions for IAS and TAS to coincide: standard conditions at sea level and negligible static source error. (L2, PDF p. 474)
- 369. c** Air density is the fourth correction listed for deriving TAS from IAS. (L2, PDF p. 476)
- 370. a** Above a certain altitude the Mach limit MMO takes over from the dynamic pressure limit VMO. (L3, PDF p. 470)
- 371. a** This page prints that the low altitude limit is a dynamic pressure limit, and PDF page 468 prints that the indicated airspeed is a dynamic pressure indicator, so the limit can be marked as a speed. (L3, PDF p. 470)
- 372. b** VMO holds or rises a little with altitude while MMO falls, because the speed of sound falls. (L3, PDF p. 470)
- 373. a** The plain red mark is the never exceed speed and the red-white mark is the autorotation limit. (L3, PDF p. 472)
- 374. b** Falling air density with altitude widens the gap between IAS and TAS. (L3, PDF p. 474)
- 375. a** The printed condition is both standard conditions at sea level and a negligible static source error, so failing one of them is enough to separate the two speeds. (L3, PDF p. 474)
- 376. b** Without an air data computer the corrections come from tables in the Flight Operations Manual. (L3, PDF p. 476)

Machmeters and vertical speed indicators

PDF pages 478–490 • 40 questions

LEVEL 1 — RECALL

- 377.** What is the purpose of the mach meter ●
- To enable flight at the never exceed speed and with maximum rate of climb
 - To enable flight at the maximum operating velocity and with minimum structural load
 - To enable flight at the maximum operating mach number and with economic efficiency
- 378.** A technician opens a mach meter on the bench. Which two sensing elements are inside it? ●
- Aneroid bellows and a capillary restrictor
 - Differential pressure bellows and a bimetallic element
 - Differential pressure bellows and aneroid bellows
- 379.** The mach number M is defined as the relation of which two speeds ●
- Speed flown to speed of sound
 - Speed of sound to speed flown
 - True airspeed to calibrated airspeed
- 380.** Which errors is the indicated mach number M_i subject to ●
- Instrument error and static source error
 - Static source error and density error
 - Instrument error and compressibility error
- 381.** In a jet aircraft, how is the speed limit shown in the airspeed indicator ●
- A red and white VMO / MMO pointer
 - A red and white VNE pointer
 - A red and green VMO / MMO pointer
- 382.** On the speed tape of a PFD, how is the speed limit shown ●
- An amber and black area
 - A red and black area
 - A red and white area
- 383.** What does the variometer measure in order to display vertical speed ●
- Vertical acceleration forces
 - Air pressure changes
 - Air temperature changes
- 384.** In which units is the variometer scale calibrated ●
- Knots or m/sec
 - Ft/min or m/sec
 - Ft/sec or m/min
- 385.** What does the abbreviation IVSI stand for ●
- Instantaneous Vertical Speed Indicator
 - Integrated Vertical Speed Indicator
 - Inertial Vertical Speed Indicator

- 386.** What does the IVSI indicator use to produce its undelayed indication ●
- A small fan with a spring loaded vane
 - A small pump with a magnetically held piston
 - A small pump with a spring loaded piston
- 387.** A diaphragm variometer is stripped for overhaul. What seals the air gap? ●
- A felt washer
 - Badger hair
 - A rubber lip seal
- 388.** What sets the central position of a diaphragm variometer ●
- A permanent magnet
 - A coiled spring
 - A hairspring driven by the capillary

LEVEL 2 — UNDERSTAND AND EXPLAIN

- 389.** Why is a mach indication needed for flight at high altitude ●
- Because the limit there is the maximum operating velocity and dynamic pressure is a function of altitude
 - Because the limit there is the maximum operating mach number and the speed of sound is a function of dynamic pressure
 - Because the limit there is the maximum operating mach number and the speed of sound is a function of altitude
- 390.** What does the shock formation caused by compression of the air do to the airflow and the forces on the aircraft ●
- It prevents turbulent airflow, decreases drag and increases lift
 - It prevents laminar airflow, increases drag and decreases lift
 - It prevents laminar airflow, decreases drag and decreases lift
- 391.** Why can the speed of sound be represented as a function of altitude ●
- Because it is a function of temperature, and under standard conditions temperature is a function of altitude
 - Because it is a function of density, and under standard conditions density is a function of airspeed
 - Because it is a function of static pressure, and under standard conditions pressure is a function of temperature
- 392.** The MMO marking on a stand-alone mach meter is the adjustable version and has to be re-set. Where is this carried out?
- On the ramp before each flight
 - In the workshop
 - In the cockpit by the pilot
- 393.** How is the indicated mach number M_i obtained compared with the true mach number M_T ●
- M_i is calculated from pitot pressure and dynamic pressure while M_T is calculated from the speed of sound and the outside temperature
 - M_i is calculated by the air data computer while M_T is calculated from pitot pressure and dynamic pressure
 - M_i is calculated from pitot pressure and dynamic pressure while M_T is calculated by the air data computer

394. From which pressures is the indicated mach number M_i calculated ●
- Pitot pressure and accumulator pressure
 - Dynamic pressure and cabin pressure
 - Pitot pressure and dynamic pressure
395. Which computer calculates the true mach number M_T ●
- The flight management computer
 - The central maintenance computer
 - The air data computer
396. Two mach values are available on an aircraft, one free of instrument and static source error. Which value is that? ●
- The maximum operating mach number MMO
 - The true mach number M_T
 - The indicated mach number M_i
397. Why do jet aircraft need a pointer or speed tape marking to show the maximum allowed airspeed
- Because the speed limit is a fixed value as the VNE is in piston engine aircraft
 - Because the speed limit is not a fixed value as the VNE is in piston engine aircraft
 - Because the airspeed indicator cannot show indicated airspeed above the speed of sound
398. How does VMO behave with increasing altitude when the compressibility error of the IAS is taken into account ●
- It always decreases in the same way as MMO
 - It is either constant or it increases slightly
 - It is either constant or it decreases slightly
399. An aircraft climbs steadily and the MMO limit falls. Why does it fall? ●
- Because the speed of sound decreases
 - Because the compressibility error increases
 - Because the dynamic pressure decreases
400. Why do all jet aircraft have an independent aural overspeed warning ●
- Because of the danger of aircraft damage during an overspeed
 - Because the static source error becomes too large at high altitude
 - Because the airspeed indicator is unreadable at high mach numbers
401. What does the mach - airspeed warning switch contain
- The same aneroid capsule stack as an altimeter
 - The same capillary tube as a variometer
 - The same bellows as a mach meter
402. How is the aural overspeed warning produced ●
- From the cockpit loudspeakers triggered by the master warning system, or by a separate system with a horn from the stall warning
 - From the cockpit loudspeakers triggered by the air data computer, or by a separate system with a clacker sound
 - From the cockpit loudspeakers triggered by the master warning system, or by a separate system with a clacker sound

- 403.** Why is a vertical speed indicator connected to the static system only ●
- Because it works from the difference between total pressure and static pressure
 - Because it works from changes of static pressure acting on its bellows
 - Because it works from the delay of the total pressure through a capillary
- 404.** Why is the outside of the variometer bellows fed through a capillary ●
- So that the same pressure always acts inside and outside the bellows
 - So that the pressure change inside the bellows is delayed
 - So that the pressure change outside the bellows is delayed
- 405.** How are the two sides of the differential pressure bellows of a mechanical variometer supplied ●
- The inside directly with total pressure, the outside delayed via a capillary
 - The inside directly with static pressure, the outside delayed via a capillary
 - The outside directly with static pressure, the inside delayed via a capillary
- 406.** An aircraft holds a constant altitude so static pressure stays constant. What does the mechanical variometer show? ●
- Zero, because the same pressure acts inside and outside the bellows
 - A small climb, because the capillary keeps the outside pressure higher
 - A small descent, because the bellows remain partly compressed
- 407.** An aircraft with an IVSI enters and leaves a turn with a high bank angle. What should be expected? ●
- A total loss of indication, because the piston is locked by the acceleration
 - A correct vertical speed indication, because the piston compensates for bank angle
 - An inaccurate vertical speed indication, because any vertical acceleration affects the piston
- 408.** What produces the force that moves the diaphragm and the pointer of a diaphragm variometer
- The difference between static pressure and accumulator pressure
 - The difference between accumulator pressure and cabin pressure
 - The difference between static pressure and total pressure
- 409.** How is pressure balanced out in a diaphragm variometer ●
- Via a calibrated orifice in the centre of the diaphragm
 - Via the air gap between the diaphragm and the casing
 - Via a capillary tube between the diaphragm and the casing
- 410.** A precision variometer is required for a glider. Why is the diaphragm type chosen? ●
- Because diaphragm variometers need no static source
 - Because diaphragm variometers are exact
 - Because diaphragm variometers have the longest time constant

LEVEL 3 — APPLY AND REASON

- 411.** A machmeter is opened and found to contain a differential pressure bellows and an aneroid bellows. Which instruments use these same two sensing elements? ●
- The airspeed indicator uses a differential pressure capsule and the vertical speed indicator an aneroid capsule
 - The airspeed indicator uses a differential pressure capsule and the altimeter an aneroid capsule
 - The altimeter uses a differential pressure capsule and the airspeed indicator an aneroid capsule

- 412.** A jet aeroplane in high altitude cruise triggers its overspeed warning. Which limit has been exceeded, and does the warning depend on the airspeed indication? ●
- MMO, and the detection is independent of the airspeed indication
 - VMO, and the detection is independent of the airspeed indication
 - MMO, and the detection is taken from the airspeed indication
- 413.** The airspeed indication in the cockpit has failed. What does this mean for overspeed detection? ●
- Detection continues, because it is always independent of the airspeed indication
 - Detection continues only while the aircraft is below the tropopause
 - Detection is lost, because it is taken from the airspeed indication
- 414.** Why is the pointer movement of a mechanical variometer delayed when the vertical speed changes? ●
- Because the change is immediate inside the bellows but delayed outside by capillary action and the larger volume
 - Because the change is immediate outside the bellows but delayed inside by capillary action and the larger volume
 - Because the pointer linkage is deliberately damped by a friction brake
- 415.** How does a mechanical variometer respond to a change of vertical speed compared with an instantaneous vertical speed indicator? ●
- Delayed in both, because both feed the outside of the bellows through a capillary
 - Undelayed in the mechanical variometer, and delayed in the IVSI because an acceleration piston is installed
 - Delayed in the mechanical variometer, and undelayed in the IVSI because an acceleration piston is installed
- 416.** An aircraft begins a descent. How does the IVSI piston give an immediate indication? ●
- The negative gravity force closes the capillary so that the outside static pressure changes first
 - The negative gravity force moves the piston up and raises the pressure inside the bellows before the outside static pressure changes
 - The negative gravity force moves the piston down and lowers the pressure inside the bellows before the outside static pressure changes

Answers — Machmeters and vertical speed indicators

- 377. c** The manual gives exactly these two purposes for the mach meter. (L1, PDF p. 478)
- 378. c** The manual lists these two elements as the contents of a mach meter. (L1, PDF p. 478)
- 379. a** Mach number is speed flown divided by speed of sound. (L1, PDF p. 480)
- 380. a** These are the two errors stated for Mi. (L1, PDF p. 480)
- 381. a** The manual names the red and white VMO / MMO pointer. (L1, PDF p. 484)
- 382. b** The PFD speed tape uses a red and black area. (L1, PDF p. 484)
- 383. b** The variometer works from changes in air pressure. (L1, PDF p. 486)
- 384. b** The manual gives ft/min or m/sec. (L1, PDF p. 486)
- 385. a** IVSI is the instantaneous vertical speed indicator, giving undelayed indication. (L1, PDF p. 488)
- 386. c** The manual describes a small pump with a spring loaded piston. (L1, PDF p. 488)
- 387. b** The manual states the gap is sealed with badger hair. (L1, PDF p. 490)
- 388. b** A coiled spring sets the central position. (L1, PDF p. 490)
- 389. c** PDF page 470 prints that at higher altitudes the limit is the maximum operating mach number, and this page prints that the speed of sound is a function of altitude, so the limit cannot be read from an airspeed alone. (L2, PDF p. 478)
- 390. b** The page states shock formation stops laminar flow, raises drag and lowers lift. (L2, PDF p. 478)

- 391. a** The chain given is speed of sound to temperature to altitude. (L2, PDF p. 478)
- 392. b** The manual calls adjusting the MMO marking a workshop task. (L2, PDF p. 478)
- 393. c** Mi comes from the pressures at the instrument while MT is worked out by the air data computer. (L2, PDF p. 480)
- 394. c** The manual names pitot pressure and dynamic pressure as the inputs for Mi. (L2, PDF p. 480)
- 395. c** The manual states the air data computer calculates MT. (L2, PDF p. 480)
- 396. b** MT is defined as being without instrument and static source error, unlike Mi. (L2, PDF p. 480)
- 397. b** The limit varies, so it must be displayed, unlike a fixed VNE. (L2, PDF p. 482)
- 398. b** The manual states VMO is constant or rises slightly with altitude. (L2, PDF p. 482)
- 399. a** The falling speed of sound is the stated reason. (L2, PDF p. 482)
- 400. a** The stated reason is the risk of damaging the aircraft. (L2, PDF p. 484)
- 401. c** The switch uses mach meter bellows. (L2, PDF p. 484)
- 402. c** These are the two arrangements given for the aural warning. (L2, PDF p. 484)
- 403. b** The instrument senses air pressure changes with static pressure on both sides of its bellows, so it needs no pitot connection. (L2, PDF p. 486)
- 404. c** The capillary holds back the pressure change on the outside, which is what creates the difference the pointer shows. (L2, PDF p. 486)
- 405. b** Static pressure reaches the inside directly and the outside through the capillary. (L2, PDF p. 486)
- 406. a** With equal pressures either side of the bellows the pointer reads zero. (L2, PDF p. 486)
- 407. c** Any vertical acceleration acts on the piston, so turn entry and exit give a wrong reading. (L2, PDF p. 488)
- 408. a** The driving difference is between static and accumulator pressure. (L2, PDF p. 490)
- 409. b** Balancing takes place through the air gap between diaphragm and casing. (L2, PDF p. 490)
- 410. b** Their accuracy is the stated reason for their use as precision instruments. (L2, PDF p. 490)
- 411. b** PDF page 468 prints that the airspeed indicator has a differential pressure capsule and PDF page 454 that the altimeter's measuring component is an aneroid capsule, which are the two bellows listed here. (L3, PDF p. 478)
- 412. a** PDF page 470 prints that at higher altitudes the limit is the maximum allowed mach number, and this page prints that the overspeed detection is always independent of the airspeed indication. (L3, PDF p. 484)
- 413. a** Detection is stated to be always independent of the airspeed indication. (L3, PDF p. 484)
- 414. a** The capillary and the larger outer volume delay the outside pressure change. (L3, PDF p. 486)
- 415. c** PDF page 486 prints that the pointer movement is delayed on a change of vertical speed, and this page prints that the IVSI removes that delay with an acceleration piston. (L3, PDF p. 488)
- 416. b** The piston raises bellows pressure ahead of the real static pressure change. (L3, PDF p. 488)